Tech pack: CORRIE F rev 3 (door D2) - 5200 cu in hunting pack bag

Revision 3, door D2, crown kit M shown (S / L cut lists in cutting_plan.csv). Issued 20261113. Generated by designs/pack_5200/mfg_corrie_f_rev3.py from the pattern module; patterns, DXF and markers in ../patterns/. Units mm, N, g (imperial alongside).

Item	Value
Pieces	75 piece types; 77 fabric / sheet pieces; 73 webbing / cord / elastic
Ledger reconciliation	93 rows, 0 unmatched, 0 pieces without a row
Mass from pattern areas	567.2 g vs ledger 567.7 g (-0.08 %, limit 3 %); bag ledger base 1065.2 g / as hunted 1336.8 g / meat trip 1532.9 g (team B basis); primary fabrics +94.9 g
Operations	92 ops; beginner 75.2 h; factory SAM 512 min
Cost a bag	materials $259 primary / $259 alternate; + CMT: US $802, Asia $336 (ASSUMPTIONS)
Load points	23; lowest SF 1.93 (B-17 IM3 tab perimeter (double row)); every load point is released by a proof test (section 7)
HOLD	4.07 B-11 lash tabs (8)
Volatile (re-run on BL-033)	BL-033 B-22 x; BL-033 F-01; BL-033 FP width; BL-033 hinge window vs B-18; BL-033 ledger; BL-033 ledger (US-chan)
Fabric sets	PRIMARY = today's fabrics (build this). ALTERNATE = team B set (ULTRAGrid body, 70D collar, ULTRA 200X floor) only after gates ALT-1..3 and FS-O5.
Owner rules	No hook-and-loop; coil zips; mostly solid colours, camo on accents only; width <= 394 mm (15.5 in); 85 L +/- 3.

Contents: 1 cover; 2 equipment and stitch standards; 3 flats / piece maps; 4 BOM; 5 cutting plan; 6 operation sequence; 7 load points and sketches; 8 QC, release tests, defects; 9 packaging; 10 costing; 11 delta register; 12 assumptions.

2. Equipment and stitch standards

ID	Machine / tool
WF1	Single-needle compound-feed walking foot lockstitch, Juki DU-1181N (or LU-1508N). Prototype: any walking foot that runs Nm 110 + tex 90 cleanly (e.g. a portable walking-foot machine).
WF2	Heavy single-needle walking foot lockstitch, large hook, Juki DNU-1541S (or LU-1561/2210 class): webbing stacks, Hypalon, tex 135 box-X. Prototype: WF1 hand-cranked through the thick spots.
BT	Computer bartacker Juki LK-1900BN (ISO 301 path / 304 zigzag, programmable, 40 x 30 mm field): every 25 mm box-X and the slit bartacks. Prototype: box-X on WF2 with the J1 template.
PS	Programmable pattern sewer Juki AMS-210EN (220 x 100 mm field) with a webbing clamp: the 38 mm box-X that do not fit the LK-1900 field (HE, T3, B-24). Prototype / small shop: WF2 with the J1 template.
ZZ	Zigzag lockstitch (ISO 304), Juki LZ-2290 class or a portable straight / zigzag walking foot: elastic laps.
LASER	CO2 laser cutter 60-100 W, bed >= 600 x 900 mm, fume extraction: Hypalon, TPU-840D, B-12, port bindings, HDPE <= 0.8 mm (or shear + round the corners).
HK	Hot knife, blade type, on a steel / glass plate with the J8 length stop: webbing, elastic, cord.
SK	Straight knife, 5 in blade (Eastman Blue Streak II class) on a cutting table for production lays. Prototype: 45 / 60 mm rotary cutter, 600 x 900 mat, 6 x 24 in ruler.
DIE	Clicker press + steel-rule dies: only above about 500 bags a year (Hypalon tabs, B-12, keepers).
TAPE	Hot-air seam-tape machine: NOT used. Rev 3 has no taped seams (ASSUMPTION NO-TAPE: the bag is water resistant, not waterproof; contents go in the user's dry bag). Needed only for a waterproof variant.

Use	Thread (record)	Needle	SPI (mm)	Stitch
40D-100D, mesh, silpoly	bonded_nylon_tex70	Nm 90/14 DPx17, round point R / Nm 90/14 DPx17, medium ball SUK (knit mesh)	10 (2.5)	301
body seams, zips, Hypalon perimeters	bonded_nylon_tex90	Nm 110/18 DPx17, round point R	8 (3.2)	301
load box-X	bonded_nylon_tex135	Nm 140/22 DPx17, round point R	7 (3.6)	301 path (LK-1900 / AMS-210EN programmed)
bartacks, elastic laps	bonded_nylon_tex90	Nm 110/18 DPx17, round point R	LK-1900 std	304

Round point (R) or slim set point on every coated fabric, laminate, Hypalon and webbing. Never a leather / cutting point (LR, S, DI): it cuts the reinforcement and the coating and drops seam strength. Ball point (SUK) only on the power mesh. New needle every bag (prototype) or every 8 h (production).

Seam efficiency: a 301 seam is taken at 80% of the weaker strip (clothsim/fabrics.yaml, attach_system.SEAM_EFF); box-X capacity = 0.5 x 1.5 x thread break / stitch length x path (attach_system.box_x, ETA_BOX 0.5 ASSUMPTION).

Code	Seam construction
SSa2	SSa plain seam, 12 mm allowance, then both allowances turned to one side and caught by a 2nd row at 6 mm (ISO 4916 1.01.01, two rows; no binding, see ASSUMPTION NO-BIND)
SSa	SSa plain seam, single row (ISO 4916 1.01.01)
SSa-ins	SSa with a webbing tab / loop inserted between the plies (ISO 4916 class 1 with insert); tab ends reach the cut edge, a backtack each side of the tab
LSd	LSd lapped patch: piece laid flat on the panel, perimeter row(s) (ISO 4916 class 2, lapped)
EFa	EFa edge turned once and edge-stitched (ISO 4916 class 6, edge finish)
EFb	EFb edge turned twice, stitched (ISO 4916 class 6); hem round a stiffener strip
ZIP	zip set: each tape laid face down on its panel edge, sewn at 5 mm, turned and edge-stitched at 3 mm (ISO 4916 class 1, two rows per tape)
BOX	box-X: rectangle + both diagonals through the webbing stack (ISO 4915 301 path; ISO 4916 class 7 when it fixes the webbing to a panel)
BT	bartack, ISO 4915 304 zigzag lockstitch, LK-1900 standard pattern
DART	dart: fold on the dart centre, sew leg to leg, tie off at the apex (ISO 4916 1.01.01)

Jig	What
J1	Box-X templates, clear acrylic 2 mm, one per size in the load table (21x20, 21x25, 25x25, 34x20, 34x30, 34x34), slots for chalk on the rectangle and both diagonals; production: LK-1900 / AMS-210EN programs of the same paths

Jig	What
J2	Swell window corner template + clip board (8 swells, notch to notch)
J3	B-27 port gauge: 6 mm turn, 20 mm lap, facing line; a slotted ruler
J4	Webbing fold gauge: aluminium angle with stops at 15 / 30 / 45 / 50 / 60 / 100 mm
J5	LK-1900 / AMS-210EN work clamps for 25 and 38 mm webbing stacks (Hypalon + panel)
J6	Door zip board: B-02door + B-02u laid flat with the corner-tangent notches pinned to the board
J7	Lid dart template (B-29a, 4 darts, apex drill marks)
J8	Hot-knife length fence with a scale in mm and in (cut lengths in cutting_plan.csv)

3. Flats and piece maps (from the pattern pieces)

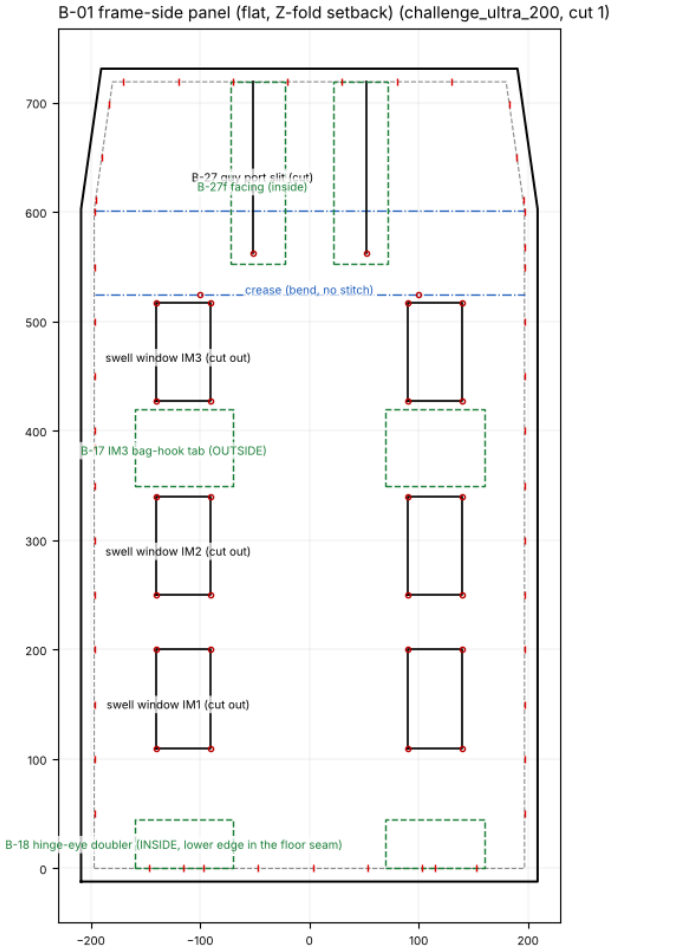


Side profile, 6-day load (45 L)

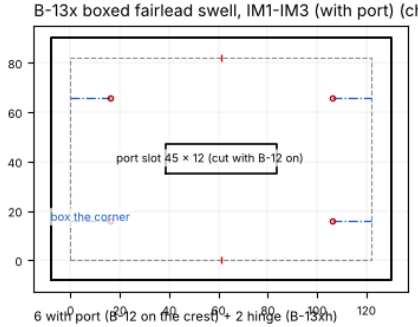
CORRIE F rev 3 (D2), H1 frame, belt B. Crown straps over the lid. Rendered 3D figure, not a photograph

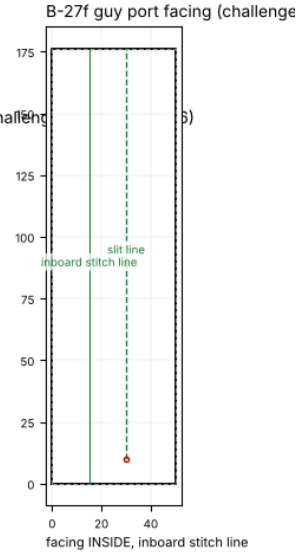
Render of rev 3 (docs/screens/20261112_BL030_2.png, BL-030): side view, lid and crown straps. Colours / camo placement as drawn there.

SK-3 B-01 frame-side panel: flat work in order 4.02 swells, 4.03 B-27 ports, 4.04 B-17 (outside), 4.05 B-18 + HE (inside), 4.06 I2K



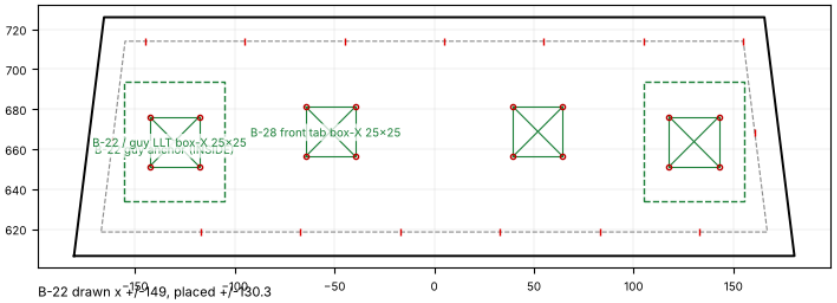
6 swell windows (cut-outs), B-27 slits x +/-52 z 595-745. Volatile: hinge window vs B-18 (BL-033).





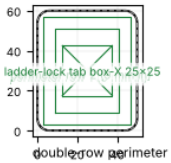
SK-4 Hinge band B-02h: B-22 INSIDE at x +/-130.3 (volatile), LLT box-X, B-28 front tab box-X (outside); then the door hinge seam

B-02h D2 hinge band (z 650 to the collar seam) (cordura_330d_nylon66, cut 1)



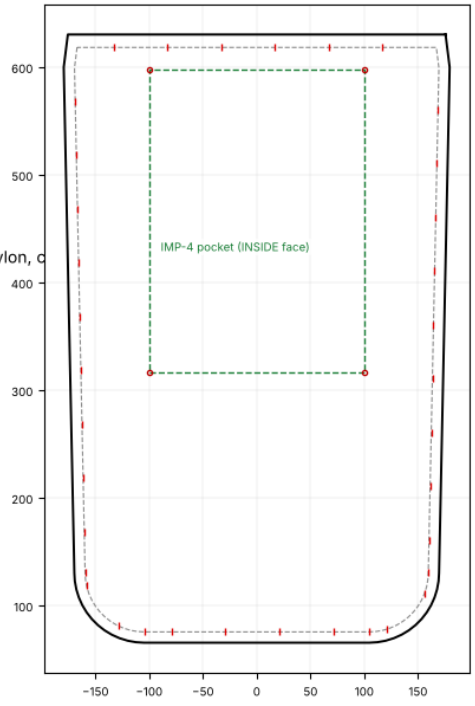
B-22 drawn x +/-149, placed +/-130.3

B-22 guy corner tab (hypalon_coated_nylon, c



double-row perimeter

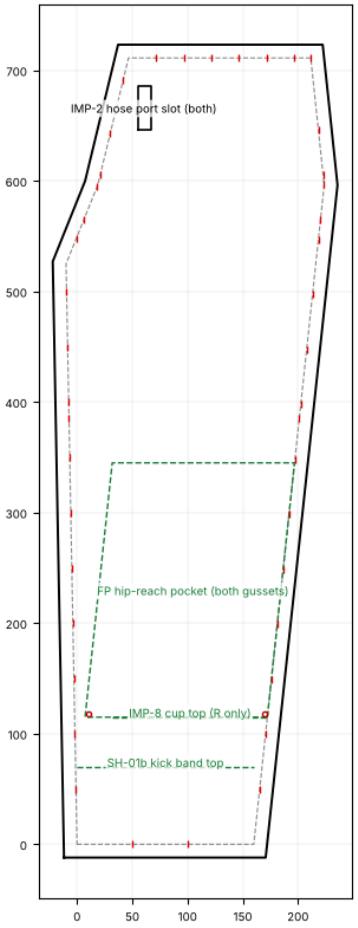
B-02door D2 door (whole aft face z 110-650, R 55 corners) (cordura_330d_nylon



IMP-4 placement INSIDE

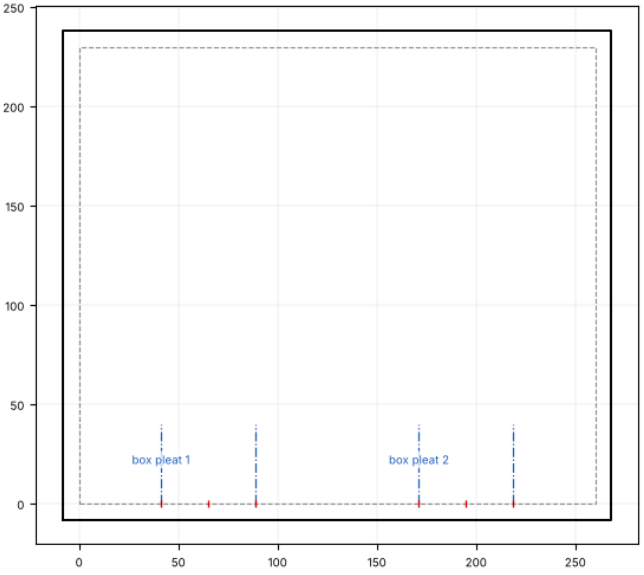
SK-5 Side gusset B-03 (cut L+R): kick band, FP pocket (volatile width), IMP-8 (R only), hose slot; corner seams carry HPL (L) and E5 (R)

B-03 side gusset (cordura_330d_nylon66, cut 2)



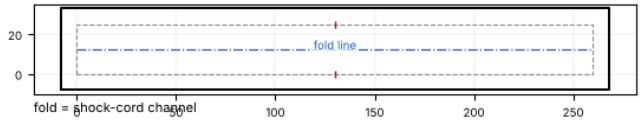
FP z 150-380

FP-a hip-reach pocket body (power mesh) (power_mesh_nylon_spandex, cut 2)



flat 260 wide, 2 box pleats

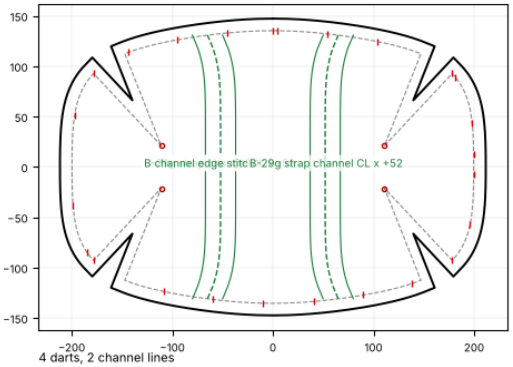
FP-b hip-reach pocket mouth band (camo) (polyester_ht_630d_printable, cut 2)



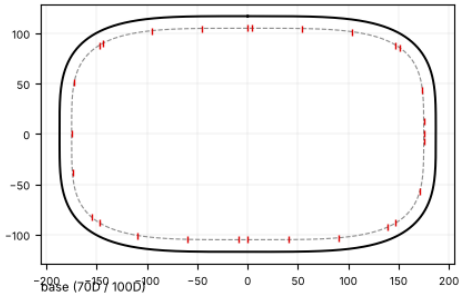
fold = shock-cord channel

SK-6 B-29 crown lid: darts → channels → zip between the aft strips → caps → crown|wall ring → wall|base ring (turn through the zip)

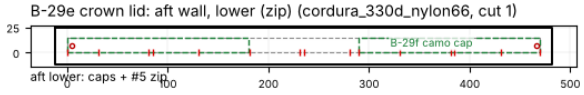
B-29a crown lid: crown (4 corner darts) (cordura_330d_nylon66, cut 1)



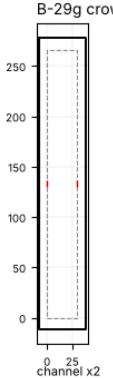
B-29b crown lid: base (lies on the roll) (robic_100d_ripstop, cut 1)



B-29e crown lid: aft wall, lower (zip) (cordura_330d_nylon66, cut 1)



B-29g crown lid: strap channel (co



4. Bill of materials (one bag, full build, crown kit M)

Production qty = 10-bag marker + 3 % lay loss where part A nested one, else net / utilisation (ASSUMPTION). Prototype (1-bag marker) quantity in the last column. Waste % = gross / net - 1.

Section	Item	Material id	Qty	Unit	Imperial	Waste %	$ / unit	$	Alternate / kit	Proto qty	Source
fabric	challenge_ultra_200: B-01x1 B-06fx1 B-13xx6 B-27fx2	challenge_ultra_200	0.487	m	0.53 yd	14.2	65.6	31.94	same / base	0.743	record challenge_ultra_200; net 0.606 m2 from the pattern areas; 10-bag marker eff 0.92 + 3 % lay loss (part A WASTE_PROD)
fabric	cordura_330d_nylon66: B-03x2 B-02doorx1 B-02hx1 B-07x1 IMP8-ax1 F-02x1 B-29ax1 B-29cx1 B-29dx1 B-29ex1 B-29gx2 A1-ax1	cordura_330d_nylon66	0.824	m	0.90 yd	23.7	15.3	12.61	cordura_330d_nylon66 / ultragrid_210d_uhmwpe_ripstop / addon,as_hunted,base	1.135	record cordura_330d_nylon66; net 1.015 m2 from the pattern areas; 10-bag marker eff 0.88 + 3 % lay loss (part A WASTE_PROD)
fabric	cordura_500d_nylon66: B-05x1 B-16x2	cordura_500d_nylon66	0.092	m	0.10 yd	34.6	16.4	1.50	challenge_ultra_200 / base	0.22	record cordura_500d_nylon66; net 0.104 m2 from the pattern areas; 10-bag marker eff 0.81 + 3 % lay loss (part A WASTE_PROD)
fabric	hdpe_sheet: B-07sx2 MS-sx2	hdpe_sheet	0.047	m2	0.51 ft2	42.9	4.58	0.21	same / base,meat_trip	0.052	record hdpe_sheet; net 0.033 m2 from the pattern areas; ASSUMPTION UTIL-SMALL 0.70 (small / sheet pieces nested across bags)
fabric	hypalon_coated_nylon: B-17x2 B-18x2 B-22x2 B-11x8	hypalon_coated_nylon	0.055	m2	0.59 ft2	42.9	27.9	1.53	same / base	0.096	record hypalon_coated_nylon; net 0.038 m2 from the pattern areas; ASSUMPTION UTIL-SMALL 0.70 (small / sheet pieces nested across bags)
fabric	polyester_ht_630d_printable: B-02ux1 SH-07x1 SH-01bx2 FP-bx2 B-29fx2 A1-dx1	polyester_ht_630d_printable	0.305	m	0.33 yd	127.6	19.7	6.00	same / addon,as_hunted,base	0.738	record polyester_ht_630d_printable; net 0.201 m2 from the pattern areas; 10-bag marker eff 0.47 + 3 % lay loss (part A WASTE_PROD)
fabric	power_mesh_nylon_spandex: FP-ax2	power_mesh_nylon_spandex	0.129	m	0.14 yd	42.9	10.9	1.41	same / base	0.246	record power_mesh_nylon_spandex; net 0.136 m2 from the pattern areas; ASSUMPTION UTIL-SMALL 0.70 (small / sheet pieces nested across bags)
fabric	robic_100d_ripstop: B-06x1 B-09cx1 IMP4-ax1 IMP4-bx1 B-29bx1 A1-bx1 MS-bx1 MS-cx1	robic_100d_ripstop	0.958	m	1.05 yd	17.8	17.5	16.77	nylon_ripstop_70d / robic_100d_ripstop / addon,as_hunted,base,meat_trip	1.208	record robic_100d_ripstop; net 1.239 m2 from the pattern areas; 10-bag marker eff 0.90 + 3 % lay loss (part A WASTE_PROD)
fabric	silicone_coated_poly_ripstop_40d: F-01ax1 F-01bx2 MS-ax1	silicone_coated_poly_ripstop_40d	1.404	m	1.54 yd	42.9	10.9	15.36	same / as_hunted,meat_trip	1.715	record silicone_coated_poly_ripstop_40d; net 1.498 m2 from the pattern areas; ASSUMPTION UTIL-SMALL 0.70 (small / sheet pieces nested across bags)
fabric	tpu_coated_nylon_840d: B-12x6 B-09px2 IMP8-bx1	tpu_coated_nylon_840d	0.036	m2	0.39 ft2	42.9	16	0.58	same / base	0.09	record tpu_coated_nylon_840d; net 0.025 m2 from the pattern areas; ASSUMPTION UTIL-SMALL 0.70 (small / sheet pieces nested across bags)
fabric_alternate	challenge_ultra_200: B-01x1 B-05x1 B-06fx1 B-13xx6 B-27fx2 B-16x2	challenge_ultra_200	0.587	m	0.64 yd	17.6	65.6	38.52	challenge_ultra_200 / cordura_500d_nylon66 / base	0.743	record challenge_ultra_200; net 0.710 m2 from the pattern areas; ASSUMPTION UTIL-ALT 0.85
fabric_alternate	cordura_330d_nylon66: IMP8-ax1 F-02x1	cordura_330d_nylon66	0.056	m	0.06 yd	42.9	15.3	0.85	same / base	0.176	record cordura_330d_nylon66; net 0.059 m2 from the pattern areas; ASSUMPTION UTIL-SMALL 0.70 (small / sheet pieces nested across bags)
fabric_alternate	hdpe_sheet: B-07sx2 MS-sx2	hdpe_sheet	0.047	m2	0.51 ft2	42.9	4.58	0.21	same / base,meat_trip	0.052	record hdpe_sheet; net 0.033 m2 from the pattern areas; ASSUMPTION UTIL-SMALL 0.70 (small / sheet pieces nested across bags)
fabric_alternate	hypalon_coated_nylon: B-17x2 B-18x2 B-22x2 B-11x8	hypalon_coated_nylon	0.055	m2	0.59 ft2	42.9	27.9	1.53	same / base	0.096	record hypalon_coated_nylon; net 0.038 m2 from the pattern areas; ASSUMPTION UTIL-SMALL 0.70 (small / sheet pieces nested across bags)

Section	Item	Material id	Qty	Unit	Imperial	Waste %	$ / unit	$	Alternate / kit	Proto qty	Source
fabric_alternate	nylon_ripstop_70d: B-06x1 B-09cx1 B-29bx1 A1-bx1 MS-bx1 MS-cx1	nylon_ripstop_70d	0.892	m	0.98 yd	17.6	11.8	10.48	robic_100d_ripstop / addon,as_hunted,base,meat_trip	1.129	record nylon_ripstop_70d; net 1.117 m2 from the pattern areas; ASSUMPTION UTIL-ALT 0.85
fabric_alternate	polyester_ht_630d_printable: B-02ux1 SH-07x1 SH-01bx2 FP-bx2 B-29fx2 A1-dx1	polyester_ht_630d_printable	0.158	m	0.17 yd	17.6	19.7	3.10	same / addon,as_hunted,base	0.738	record polyester_ht_630d_printable; net 0.201 m2 from the pattern areas; ASSUMPTION UTIL-ALT 0.85
fabric_alternate	power_mesh_nylon_spandex: FP-ax2	power_mesh_nylon_spandex	0.129	m	0.14 yd	42.9	10.9	1.41	same / base	0.246	record power_mesh_nylon_spandex; net 0.136 m2 from the pattern areas; ASSUMPTION UTIL-SMALL 0.70 (small / sheet pieces nested across bags)
fabric_alternate	robic_100d_ripstop: IMP4-ax1 IMP4-bx1	robic_100d_ripstop	0.115	m	0.13 yd	42.9	17.5	2.02	same / base	0.296	record robic_100d_ripstop; net 0.123 m2 from the pattern areas; ASSUMPTION UTIL-SMALL 0.70 (small / sheet pieces nested across bags)
fabric_alternate	silicone_coated_poly_ripstop_40d: F-01ax1 F-01bx2 MS-ax1	silicone_coated_poly_ripstop_40d	1.157	m	1.27 yd	17.6	10.9	12.65	same / as_hunted,meat_trip	1.715	record silicone_coated_poly_ripstop_40d; net 1.498 m2 from the pattern areas; ASSUMPTION UTIL-ALT 0.85
fabric_alternate	tpu_coated_nylon_840d: B-12x6 B-09px2 IMP8-bx1	tpu_coated_nylon_840d	0.036	m2	0.39 ft2	42.9	16	0.58	same / base	0.09	record tpu_coated_nylon_840d; net 0.025 m2 from the pattern areas; ASSUMPTION UTIL-SMALL 0.70 (small / sheet pieces nested across bags)
fabric_alternate	ultragrid_210d_uhmwpe_ripstop: B-03x2 B-02doorx1 B-02hx1 B-07x1 B-29ax1 B-29cx1 B-29dx1 B-29ex1 B-29gx2 A1-ax1	ultragrid_210d_uhmwpe_ripstop	0.75	m	0.82 yd	17.6	21.9	16.40	cordura_330d_nylon66 / addon,as_hunted,base	0.963	record ultragrid_210d_uhmwpe_ripstop; net 0.956 m2 from the pattern areas; ASSUMPTION UTIL-ALT 0.85
webbing	elastic_25mm: K-28 K-G K-IM K-RT	elastic_25mm	1.04	m	1.14 yd	5	0.6	0.62	same / as_hunted,base		ASSUMPTION ELASTIC-COST, no record
webbing	elastic_6mm: F-EL	elastic_6mm	2.558	m	2.80 yd	5	0.25	0.64	same / as_hunted		ASSUMPTION ELASTIC-COST, no record (fly hem)
webbing	poly_webbing_25mm: B-25 B-28 B-28t BRL BRS E5 GBL HPL I2K I2T IM1 IM2 IM3 K-W LLT RT TG US W1 W2 W3 WX	poly_webbing_25mm	16.549	m	18.10 yd	5	0.7	11.58	same / as_hunted,base,crown kit M,crown kit M (as hunted),meat_trip		record poly_webbing_25mm
webbing	poly_webbing_38mm: B-24 HE IME T3	poly_webbing_38mm	2.825	m	3.09 yd	5	1	2.82	same / base,meat_trip		record poly_webbing_38mm
webbing	shock_cord_3mm: B-26 E5-cord FP-cord	shock_cord_3mm	1.198	m	1.31 yd	5	0.5	0.60	same / base		record shock_cord_3mm
hardware	#10 coil zip chain, door D2 (2 sliders)	coil_zipper_10	1.418	m	1.55 yd	0	4	5.67	same / base		record coil_zipper_10; ledger rows 'D2 door zip'
hardware	#10 coil slider, metal pull (door)	slider_10	2	unit		0	0.8	1.60	same / base		ASSUMPTION SLIDER: #10 coil slider, metal pull; ledger rows 'D2 door zip'
hardware	#5 coil zip chain: lid, IMP-4, F-02 (3 zips)	coil_zipper_5	0.97	m	1.06 yd	0	2	1.94	same / base,as_hunted		record coil_zipper_5; ledger rows 'IMP-4 / add-on crown_lid / F-02'
hardware	#5 coil slider + pull	slider_5	3	unit		0	0.4	1.20	same / base,as_hunted		ASSUMPTION SLIDER: #5 coil slider; ledger rows 'IMP-4 / crown_lid'
hardware	door pull cord 3 mm Dyneema, 2 x 0.15 m (ASSUMPTION length)	dyneema_cord_3mm	0.3	m	0.33 yd	0	0.9	0.27	same / base		record dyneema_cord_3mm; ledger rows 'door: 2 cord pulls'
hardware	rubber pull sleeve	rubber_pull_sleeve	2	unit		0	0.15	0.30	same / base		ASSUMPTION; ledger rows 'door: 2 cord pulls'
hardware	rated 25 mm aluminium G-hook: IM1, IM2 (2) + W1-W3 (3) + bow riser (1)	g_hook_25mm_aluminium	6	unit		0	4	24.00	same / base		record g_hook_25mm_aluminium; ledger rows 'IM-hooks / W-hw / BL-010 S9 / bow riser'
hardware	rated 25 mm aluminium G-hook: W haul extenders	g_hook_25mm_aluminium	3	unit		0	4	12.00	same / meat_trip		record g_hook_25mm_aluminium; ledger rows 'BL-031: W1-W3 haul extenders'
hardware	wiregate carabiner 22 kN: IM3 bag hooks (2) + IM3 meat free end (1)	carabiner_wiregate_22kn	3	unit		0	10	30.00	same / base		record carabiner_wiregate_22kn; ledger rows 'BL-010 A1 / BL-010 S2'

Section	Item	Material id	Qty	Unit	Imperial	Waste %	$ / unit	$	Alternate / kit	Proto qty	Source
hardware	wiregate carabiner 22 kN: B-24 float links	carabiner_wiregate_22kn	2	unit		0	10	20.00	same / meat_trip		record carabiner_wiregate_22kn; ledger rows 'BL-010 S12'
hardware	heavy 25 mm ladder lock: B-28 (4) + guys (2) + W1-W3 (3)	ladder_lock_25mm_heavy	9	unit		0	2.1	18.90	same / base,as_hunted		record ladder_lock_25mm_heavy; ledger rows 'W-hw / BL-010 A3 / crown_straps'
hardware	25 mm aluminium cam buckle: IM1-IM3 (3) + US1/US2 (2)	cam_25mm_al	5	unit		0	6	30.00	same / base		ASSUMPTION CAM25 part (MT-3); no record; ledger rows 'IM-cams / US-cams'
hardware	25 mm tri-glide: US keepers (2) + guy keepers (2)	triglide_25mm	4	unit		0	0.3	1.20	same / base		ASSUMPTION; attach_system.TRIGLIDE_G mass; ledger rows 'US-tg / guy ladder locks'
hardware	POM toggle 30 x 8 x 5: B-28 rear (2) + guy loops (2)	pom_toggle	4	unit		0	0.3	1.20	same / base,as_hunted		ASSUMPTION; attach_system.TOGGLE_G 3 g, POM 30 x 8 x 5; ledger rows 'crown_straps / BL-004 REC guy bar loops'
hardware	25 mm side-release buckle (roll top)	buckle_side_release_25mm	2	unit		0	0.8	1.60	same / base		record buckle_side_release_25mm; ledger rows 'roll:'
hardware	spring cord lock (E-5 pole grip)	cord_lock_spring	1	unit		0	0.4	0.40	same / base		record cord_lock_spring; ledger rows 'E-5 pole quiver'
hardware	TPU sleeve on the IM3 carabiner (2)	tpu_hook_sleeve	2	unit		0	0.2	0.40	same / base		ASSUMPTION; ledger rows 'attachment: TPU hook sleeves (4) less BL-004 REC (2)'
hardware_frame_side	printed fairleads (8), 7075 pins (8), UHMWPE sleeves (8), US anchors, lifter keepers	frame kit	1	set		0			/ base		ledger rows 'attachment: printed fairleads ...'
thread	bonded_nylon_tex135 (needle + bobbin)	bonded_nylon_tex135	34.6	m	37.84 yd	15	0.0081	0.28	same / all		record bonded_nylon_tex135; sum of op stitch lengths x 2.75 (ASSUMPTION THREAD-FACTOR)
thread	bonded_nylon_tex70 (needle + bobbin)	bonded_nylon_tex70	83.6	m	91.43 yd	15	0.0052	0.43	same / all		record bonded_nylon_tex70; sum of op stitch lengths x 2.75 (ASSUMPTION THREAD-FACTOR)
thread	bonded_nylon_tex90 (needle + bobbin)	bonded_nylon_tex90	121.1	m	132.44 yd	15	0.0061	0.74	same / all		record bonded_nylon_tex90; sum of op stitch lengths x 2.75 (ASSUMPTION THREAD-FACTOR)
consumable	6 mm double-sided basting tape (zips)	basting_tape_6mm	3.66	m	4.00 yd	0	0.05	0.18	/		ASSUMPTION
consumable	labels: size / care / ID, hang tag, load-limit warning card	label_set	1	set		0	0.6	0.60	/		ASSUMPTION: size / care / ID label + hang tag + warning card
consumable	silicone seam sealer for the silpoly fly and liner seams, ~10 ml (ASSUMPTION)	seam_sealer	1	bag		0	0.8	0.80	/		ASSUMPTION: 10 ml of a silicone seam sealer
packaging	poly bags + carton share	polybag_carton	1	bag		0	0.9	0.90	/		ASSUMPTION: 2 recycled poly bags + 1/10 carton

5. Cutting plan

Markers: ../patterns/markers/. Cut windows, slots and slits first; knife clockwise from the grain arrow; notches 6 mm. Ply height: woven nylon 24, laminates / mesh / silpoly 12, camo single ply (one way, print placement), sheet goods laser single sheet.

Set	Material	Pieces	1-bag m	eff	10-bag m / bag	eff	Lay	Prototype	Production
primary	challenge_ultra_200	B-01x1 B-06fx1 B-13xx6 B-27fx2	0.743	0.5838	0.4726	0.9177	laminate face up, all pieces one way	rotary; windows and B-27 slits with a hot knife / scalpel on a cutting mat	straight knife, 12 plies max (laminate creeps), or laser single ply
primary	cordura_330d_nylon66	B-03x2 B-02doorx1 B-02hx1 B-07x1 IMP8-ax1 F-02x1 B-29ax1 B-29cx1 B-29dx1 B-29ex1 B-29gx2 A1-ax1	1.1348	0.6178	0.7996	0.8767	face up, one way not needed	rotary + hot-knife the window / slot cut-outs	straight knife, 10-bag marker, 24 plies max; or laser single ply (sealed edges)
primary	cordura_500d_nylon66	B-05x1 B-16x2	0.22	0.3255	0.0889	0.8055	face up	rotary	straight knife, 24 plies max
primary	hdpe_sheet	B-07sx2 MS-sx2	0.043	0.6213			free rotation	shear / router, round every corner R >= 3	CNC router from the DXF
primary	hypalon_coated_nylon	B-17x2 B-18x2 B-22x2 B-11x8	0.07	0.4063			coated face up, free rotation	laser from the DXF (makerspace) or a knife + template	laser single sheet; dies above 500 bags
primary	polyester_ht_630d_printable	B-02ux1 SH-07x1 SH-01bx2 FP-bx2 B-29fx2 A1-dx1	0.738	0.1878	0.2961	0.468	print face up, ONE WAY	rotary	single ply (printed camo, one way, pattern placement)
primary	power_mesh_nylon_spandex	FP-ax2	0.246	0.3807			stretch ACROSS, one way	rotary, no stretch while cutting	straight knife 12 plies, relaxed 24 h before cutting
primary	robic_100d_ripstop	B-06x1 B-09cx1 IMP4-ax1 IMP4-bx1 B-29bx1 A1-bx1 MS-bx1 MS-cx1	1.208	0.6966	0.9305	0.9043	face up	rotary	straight knife, 24 plies max (paper between every 6 plies: slippery)
primary	silicone_coated_poly_ripstop_40d	F-01ax1 F-01bx2 MS-ax1	1.7146	0.5932			face up	rotary, weights not pins	straight knife 12 plies with paper underlay
primary	tpu_coated_nylon_840d	B-12x6 B-09px2 IMP8-bx1	0.06	0.2887			coated face up, free rotation	laser from the DXF	laser single sheet
alternate	challenge_ultra_200	B-01x1 B-05x1 B-06fx1 B-13xx6 B-27fx2 B-16x2	0.743	0.6837			laminate face up, all pieces one way	rotary; windows and B-27 slits with a hot knife / scalpel on a cutting mat	straight knife, 12 plies max (laminate creeps), or laser single ply
alternate	cordura_330d_nylon66	IMP8-ax1 F-02x1	0.176	0.2327			face up, one way not needed	rotary + hot-knife the window / slot cut-outs	straight knife, 10-bag marker, 24 plies max; or laser single ply (sealed edges)
alternate	hdpe_sheet	B-07sx2 MS-sx2	0.043	0.6213			free rotation	shear / router, round every corner R >= 3	CNC router from the DXF
alternate	hypalon_coated_nylon	B-17x2 B-18x2 B-22x2 B-11x8	0.07	0.4063			coated face up, free rotation	laser from the DXF (makerspace) or a knife + template	laser single sheet; dies above 500 bags
alternate	nylon_ripstop_70d	B-06x1 B-09cx1 B-29bx1 A1-bx1 MS-bx1 MS-cx1	1.129	0.6714			face up	rotary	straight knife, 12 plies max (slippery)
alternate	polyester_ht_630d_printable	B-02ux1 SH-07x1 SH-01bx2 FP-bx2 B-29fx2 A1-dx1	0.738	0.1878			print face up, ONE WAY	rotary	single ply (printed camo, one way, pattern placement)
alternate	power_mesh_nylon_spandex	FP-ax2	0.246	0.3807			stretch ACROSS, one way	rotary, no stretch while cutting	straight knife 12 plies, relaxed 24 h before cutting
alternate	robic_100d_ripstop	IMP4-ax1 IMP4-bx1	0.296	0.2819			face up	rotary	straight knife, 24 plies max (paper between every 6 plies: slippery)
alternate	silicone_coated_poly_ripstop_40d	F-01ax1 F-01bx2 MS-ax1	1.7146	0.5932			face up	rotary, weights not pins	straight knife 12 plies with paper underlay
alternate	tpu_coated_nylon_840d	B-12x6 B-09px2 IMP8-bx1	0.06	0.2887			coated face up, free rotation	laser from the DXF	laser single sheet
alternate	ultragrid_210d_uhmwpe_ripstop	B-03x2 B-02doorx1 B-02hx1 B-07x1 B-29ax1 B-29cx1 B-29dx1 B-29ex1 B-29gx2 A1-ax1	0.963	0.6729			face up	rotary (UHMWPE grid: new blade)	straight knife 12 plies; UHMWPE dulls blades: sharpen every lay

Webbing, cord and elastic cut lengths

ID	Use	Material	Length mm (in)	Qty	Kit	Source
B-28	B-28 crown strap, kit S	poly_webbing_25mm	1028 (40.5)	2	crown kit S (as hunted)	corrie_f.crown_kit_straps
B-28	B-28 crown strap, kit M	poly_webbing_25mm	967 (38.1)	2	crown kit M (as hunted)	corrie_f.crown_kit_straps
B-28	B-28 crown strap, kit L	poly_webbing_25mm	908 (35.7)	2	crown kit L (as hunted)	corrie_f.crown_kit_straps
B-28t	B-28 lock tab (front on the hinge band, rear through the B-25 toggle, B-25 second fold)	poly_webbing_25mm	60 (2.4)	6	as_hunted	corrie_f.CROWN_TAB, 3 per strap
K-28	B-28 tail keeper (25 mm elastic loop)	elastic_25mm (no record: ASSUMPTION)	90 (3.5)	4	as_hunted	corrie_f.crown_kit_straps keepers_per_strap x 2
W1	haul wrap W1 (cut for carry)	poly_webbing_25mm	750 (29.5)	1	base	strap_tails.wrap_rows
W2	haul wrap W2 (cut for carry)	poly_webbing_25mm	822 (32.4)	1	base	strap_tails.wrap_rows
W3	haul wrap W3, kit S	poly_webbing_25mm	1034 (40.7)	1	crown kit S	strap_tails.wrap_rows
W3	haul wrap W3, kit M	poly_webbing_25mm	987 (38.8)	1	crown kit M	strap_tails.wrap_rows
W3	haul wrap W3, kit L	poly_webbing_25mm	935 (36.8)	1	crown kit L	strap_tails.wrap_rows
WX	W haul extender (meat-trip kit)	poly_webbing_25mm	740 (29.1)	3	meat_trip	strap_tails.wrap_rows haul extender_mm
K-W	W gore keeper loop (6, in the gore seams)	poly_webbing_25mm	80 (3.1)	6	base	ledger 'keepers: 6 haul-wrap keeper loops' (ASSUMPTION 80 mm cut)
RT	roll-top side clip strap	poly_webbing_25mm	510 (20.1)	2	base	strap_tails.roll_row
K-RT	roll clip tail keeper (25 mm elastic)	elastic_25mm (no record: ASSUMPTION)	90 (3.5)	2	base	BL-031
IM1	internal meat strap IM1	poly_webbing_25mm	990 (39.0)	1	base	strap_tails.im_rows (unchanged)
IM2	internal meat strap IM2	poly_webbing_25mm	1023 (40.3)	1	base	strap_tails.im_rows (unchanged)
IM3	internal meat strap IM3	poly_webbing_25mm	1083 (42.6)	1	base	strap_tails.im_rows (unchanged)
K-IM	IM tail keeper (25 mm elastic)	elastic_25mm (no record: ASSUMPTION)	90 (3.5)	3	base	BL-031
IME	IM fairlead eye (38 mm)	poly_webbing_38mm	200 (7.9)	6	base	ledger IM-eyes 6 x 0.20 m
US	under-strap US1 / US2	poly_webbing_25mm	550 (21.7)	2	base	ledger US-web 1.10 m
B-24	float stand-off link (38 mm, meat-trip)	poly_webbing_38mm	435 (17.1)	2	meat_trip	ledger B-24-web 2 x 435 mm
HE	hinge eye (38 mm)	poly_webbing_38mm	150 (5.9)	2	base	attach_system.drawing_pieces HE
T3	IM3 bag-hook tongue (38 mm)	poly_webbing_38mm	160 (6.3)	2	base	attach_system.drawing_pieces T2/T3 (IM2 hooks removed by BL-004: 2 left)
B-25	guy bridle on the crown top bar	poly_webbing_25mm	412 (16.2)	1	base	corrie_f.guy_bridle
TG	port guy (B-25 toggle -> B-27 port -> B-22 inside)	poly_webbing_25mm	500 (19.7)	2	base	attach_system.GUY cut_mm
LLT	guy ladder-lock tab on B-22	poly_webbing_25mm	70 (2.8)	2	base	attach_system.drawing_pieces LLT
K-G	guy tail keeper (25 mm elastic)	elastic_25mm (no record: ASSUMPTION)	90 (3.5)	2	base	ledger 'guy tail elastic keepers x2'
GBL	guy bar loop (POM toggle)	poly_webbing_25mm	170 (6.7)	2	base	ledger BL-004 REC 'guy bar loops x2: 25 mm x 170 mm'
HPL	scope-pod hard-point loop	poly_webbing_25mm	80 (3.1)	4	base	corrie_f.LOOP_M
E5	pole grip tab (cord lock)	poly_webbing_25mm	60 (2.4)	1	base	evolve.POLE tab_m
I2T	IMP-2 hang tab (B-06f)	poly_webbing_25mm	100 (3.9)	1	base	lean.fell IMP-2L 0.10 m
I2K	IMP-2 bottom keeper tab (B-01)	poly_webbing_25mm	60 (2.4)	2	base	lean.fell IMP-2L 0.06 m

ID	Use	Material	Length mm (in)	Qty	Kit	Source
BRL	bow riser loop (through B-11)	poly_webbing_25mm	100 (3.9)	2	base	ledger bow riser carry (ASSUMPTION 100 mm loops)
BRS	bow riser strap + G-hook	poly_webbing_25mm	300 (11.8)	1	base	ledger bow riser carry (ASSUMPTION 300 mm strap)
B-26	B-26 tripod keeper, 3 mm shock cord (ends 15 mm in the R corner seams)	shock_cord_3mm	201 (7.9)	1	base	corrie_f.tripod_keeper
FP-cord	FP mouth shock cord	shock_cord_3mm	320 (12.6)	2	base	corrie_f.hip_reach_pockets (w + pleat + 60)
E5-cord	pole quiver grip cord	shock_cord_3mm	300 (11.8)	1	base	evolve.POLE
F-EL	fly hem elastic, 6 mm (spine top + bottom, sides bottom + frame)	elastic_6mm (no record: ASSUMPTION)	2436 (95.9)	1	as_hunted	corrie_f.fly_pieces (FLY_ELASTIC_RATIO x hem channels)

6. Sewing operation sequence

All flat work before the bag becomes 3D. Clips, never pins, in coated fabric. Swatch every new stack. Load points only after their proof test passes (section 8). V = volatile (re-run on BL-033).

Seq	Operation	How (beginner)	Pieces	Machine	Needle	Thread	SPI	Stitch / seam	mm	Ref	h / SAM min
0.01	Print and check the test square	Print patterns/print_letter/00_PRINT_FIRST_test_square.pdf at 100 %. The solid square must read 100 mm (+/- 1 mm) both ways, the dashed one 4 in. Do not cut from a wrong-scale print.		bench	-	-	-	- / -		QC-01	0.10 / 0.0 1x
0.02	Machine set-up and stack swatches	Make a 150 x 150 mm swatch of each stack and set tension until the lock sits mid-stack on both faces: (a) ULTRA + 330D body seam, tex 90 / Nm 110 R, 8 SPI; (b) Hypalon on ULTRA, double row; (c) 25 mm box-X tex 135 / Nm 140 R, 7 SPI; (d) 70D / mesh at 10 SPI. Keep them in the QC folder.		WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / SSa2, LSd, BOX, SSa (swatches)		SK-1, SK-2; QC-05, QC-06	1.00 / 0.0 1x
0.03	Sew the first-article load samples	Before cutting the bag, sew the proof specimens exactly as the bag: 10 x 25 mm box-X loops (MT-1), 5 T3 tongues on 70 mm B-17 tabs on B-01 swatches (PT-2), 5 HE eyes on B-18 + B-01 + B-05 swatches without binding (PT-3), 5 B-24 links (PT-6). The bag's load points are sewn only after these pass.		BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX + LSd (as the bag)		SK-1; TEST gates; all load points	3.00 / 0.0 1x
0.04	Cut 77 fabric / sheet pieces	Per cutting_plan.csv: lay face up (one way for camo and mesh), L+R pieces face to face, cut windows, slots and slits FIRST, knife clockwise from the grain arrow. Notches 6 mm, never past the seam line. Hypalon, TPU-840D and HDPE from the DXF on the laser / router.	all	SK / LASER / HK	-	-	-	- / -		QC-02, QC-03, QC-04	9.24 / 66.5
0.05	Hot-knife 73 webbing / elastic / cord pieces	Cut lengths in cutting_plan.csv (crown kit M shown; B-28 and W3 per crown kit). Use the J8 stop; seal both ends; no hard bead wider than the webbing.	webbing	SK / LASER / HK	-	-	-	- / -		QC-04	2.19 / 15.8
0.06	Mark every placement	Chalk / silver pen on the INSIDE unless the pattern says OUTSIDE; awl the drill marks (box corners, slit ends, window corners, zip stops). Pieces: B-01, B-03 (L only / R only marks), B-02h, B-02door, B-02u, B-06f, B-29a.		bench	-	-	-	- / -			0.60 / 4.3
1.01 V	FP hip-reach pocket: pleats	Fold the 2 box pleats on each FP-a (flat 260 mm wide, finished 165 mm), baste across top and bottom at 6 mm. Do not stretch the mesh.	FP-a x2	WF1	Nm 90/14 DPx17, medium ball SUK (knit mesh)	BN tex70	10	301 / SSa	1040	SK-5	1.02 / 7.4
1.02 V	FP mouth band and shock-cord channel	Sew FP-b to the FP-a top (260 mm), fold the band on its fold line over the seam, edge-stitch 2 mm to form the channel; thread the shock cord (FP-cord) and bar it at both ends inside the side allowance.	FP-a, FP-b x2, FP-cord	WF1	Nm 90/14 DPx17, medium ball SUK (knit mesh)	BN tex70	10	301 / SSa + EFa channel	1040	SK-2	0.62 / 4.5
1.03	IMP-4 zip pocket	Set the #5 zip (210 mm) in the top of IMP4-a (basting tape, ZIP seam), stops at the drill marks; sew the bellows strip IMP4-b round bottom + sides of IMP4-a.	IMP4-a, IMP4-b, #5 zip	WF1	Nm 90/14 DPx17, round point R	BN tex70	10	301 / ZIP + SSa	1240		2.29 / 16.5
1.04	IMP-8 cam cup	Fold IMP8-a, sew the 2 side seams; sew the laser-cut TPU base IMP8-b in (TPU face out).	IMP8-a, IMP8-b	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / SSa	610		0.75 / 5.4
1.05	IMP-2 hydration sleeve	Fold B-09c into a tube, one side seam; turn the top edge 10 mm and edge-stitch; I2T hang tab goes in later (op 5.01).	B-09c	WF1	Nm 90/14 DPx17, round point R	BN tex70	10	301 / SSa + EFa	930		0.56 / 4.0
1.06 V	F-02 fly base pocket	Set the #5 zip (230 mm), turn the edges 10 mm. It is sewn to the floor in op 4.13. Placement on B-05 is NOT drawn on the pattern: centre it on the outside of B-05 (ASSUMPTION F02-PLACE).	F-02, #5 zip	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / ZIP + EFa	1120		1.13 / 8.1
2.01	B-12 strap-port patches on the swell crests	Lay a B-12 (TPU face out) on the crest of each B-13x (6, not the hinge swells), sew the 3 mm perimeter, then cut the 45 x 12 slot through both and edge-stitch the slot.	B-12 x6, B-13x x6	WF2	Nm 110/18 DPx17, round point R	BN tex90	8	301 / LSd	1843	SK-2	1.47 / 10.6

Seq	Operation	How (beginner)	Pieces	Machine	Needle	Thread	SPI	Stitch / seam	mm	Ref	h / SAM min
2.02 V	Box the swells	Sew the 4 corner folds of every swell (B-13x x6): fold on 'box the corner', sew across. Practice: set one swell into a scrap window before touching B-01.	B-13x x6	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / SSa			1.50 / 10.8
2.03	T3 IM3 tongues onto B-17	Fold each T3 45 mm round the bed (empty loop: the carabiner clips in later), box-X the loop; box-X the other end to the centre of a B-17 (Hypalon, coated face out).	T3 x2, B-17 x2	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	902	SK-1; QC-07; T3 (load table)	0.40 / 2.9
2.04	IME fairlead eyes (frame kit)	Fold each 38 mm IME at 100 mm into an eye, box-X 34 x 30. These go on the frame fairleads, not the bag.	IME x6	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	1312	QC-07; IME	0.60 / 4.3
3.01	SH-07 slider garage on the right blade	Turn the edges 8 mm and topstitch SH-07 to B-02u at its placement.	SH-07, B-02u	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / LSd	200		0.16 / 1.1
3.02	IMP-4 onto the door, INSIDE face	Sew the IMP-4 bellows edge to B-02door at the IMP-4 placement (inside), pivot at the drill marks.	IMP-4, B-02door	WF1	Nm 90/14 DPx17, round point R	BN tex70	10	301 / SSa	820		0.69 / 5.0
3.03	Door #10 zip	Basting-tape the #10 zip to the door edge and to the blade U inner edge (1358 mm, 53.5 in), notch to notch, corner tangent notches at each R 55 corner; 2 sliders head to head; sew each tape at 5 mm, turn, edge-stitch 3 mm; zip stops at the hinge band.	B-02door, B-02u, #10 zip	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / ZIP	5432	SK-2; QC-09	2.66 / 19.1
3.04 V	B-22 guy anchors INSIDE the hinge band	Place each B-22 (coated face down on the band's inside) at x +/-130.3 mm (5.13 in) on the drawn placement (drawn x +/-149, moved 18.7 mm inboard, ASSUMPTION B22-IN). Double-row perimeter 3 / 9 mm.	B-22 x2, B-02h	WF2	Nm 110/18 DPx17, round point R	BN tex90	8	301 / LSd (2 rows)	839	SK-4; QC-07	0.67 / 4.8
3.05 V	Guy ladder-lock tabs LLT through B-22	Fold each LLT round the bar of a heavy 25 mm ladder lock, box-X through LLT + B-22 + B-02h at the drawn 'B-22 / guy LLT box' (drill corners).	LLT x2, B-22, B-02h, 2 heavy ladder locks	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	315	SK-1, SK-4; QC-07; LLT + B-22 + B-02h	0.20 / 1.4
3.06	B-28 front tabs on the hinge band (OUTSIDE)	Fold a B-28t round a heavy ladder lock, box-X it on the OUTSIDE of B-02h at the drawn 'B-28 front tab box' (ASSUMPTION B28-FACE: the strap runs on the aft face, so the tab is outside; part A README says 'inside').	B-28t x2, B-02h, 2 heavy ladder locks	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	341	SK-4; QC-07; B-28t	0.20 / 1.4
3.07	Hinge band to the door top	Join B-02h to the door top (334 mm), SSa2.	B-02h, B-02door	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / SSa2	668		0.20 / 1.4
3.08	Hinge band ends to the blade U	Join both short band ends to the blade tops (193 mm total).	B-02h, B-02u	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / SSa2	385		0.22 / 1.6
4.01 V	Check the B-01 cut-outs	6 swell windows, 2 B-27 slits (156 mm at x +/-52), IMP-2 keeper drill marks. All cut, corners awled.	B-01	bench	-	-	-	- / -		QC-03	0.10 / 0.7
4.02 V	Set the 6 swells into the B-01 windows	Swell base to window edge, INSIDE B-01, notch to notch (6 windows, 1680 mm of window edge in all; hinge windows: 0). Sew from the swell side, needle down at every corner.	B-13x x6, B-01	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / SSa	1680	SK-3; QC-08	1.50 / 10.8
4.03	B-27 guy ports (2)	Each slit (156 mm, x +/-52, z 595-745): turn both edges 6 mm and edge-stitch; sew the B-27f facing INSIDE along the inboard stitch line so the outboard edge laps 20 mm over it; bartack both slit ends. It crosses the setback crease at z 631: the facing follows the crease.	B-01, B-27f x2	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / EFa + LSd (lapped port)	965	SK-2, SK-3; QC-10; B-27 slit ends	0.98 / 7.0
4.04	B-17 IM3 tabs OUTSIDE B-01	Place each B-17 (with its T3) OUTSIDE B-01 at x +/-115, z 385-455 (placement). Double-row perimeter 3 / 9 mm, 8 SPI, tex 90; the 90 mm top edge carries the load: no gaps, backtack 10 mm.	B-17 x2, B-01	WF2	Nm 110/18 DPx17, round point R	BN tex90	8	301 / LSd (2 rows)	1239	SK-3; QC-07; B-17 perimeter	0.99 / 7.1
4.05 V	B-18 doublers + HE hinge eyes	Lay each B-18 INSIDE B-01 at the placement (lower edge to the floor seam line). Fold the HE eye (50 mm back, 25 mm eye; first 30 mm tri-folded to an 18 mm neck, 304 bartack across the neck), box-X 34 x 34 through HE + B-18 + B-01, then the B-18 perimeter rows on 3 sides (the 4th goes in the floor seam).	HE x2, B-18 x2, B-01	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX + LSd	1243	SK-1, SK-3; QC-07; HE + B-18 + B-01	1.02 / 7.4

Seq	Operation	How (beginner)	Pieces	Machine	Needle	Thread	SPI	Stitch / seam	mm	Ref	h / SAM min
4.06	IMP-2 keeper tabs I2K on B-01	Fold each I2K, bartack at the 2 'IMP-2 keeper tab' drill marks (inside).	I2K x2, B-01	BT	Nm 110/18 DPx17, round point R	BN tex90	-	304 / BT			0.20 / 1.4
4.07	B-11 lash tabs (8)	Bow riser + muzzle keepers. NOT placed on any rev 3 pattern piece (found in this ticket). HOLD until the pattern-maker draws the placements; sew-down box 301 tex 90 when placed. HOLD (pattern).	B-11 x8	WF2	Nm 110/18 DPx17, round point R	BN tex90	8	301 / LSd			0.80 / 5.8
4.08	Gusset flat work: kick bands	SH-01b on each gusset foot at the 'kick band top' line, top edge turned 8 mm, topstitched; bottom and sides go in the seams.	SH-01b x2, B-03 x2	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / LSd	320		0.26 / 1.9
4.09 V	Gusset flat work: FP pockets (both)	Sew each FP to its gusset at the placement, z 150-380, 165 mm face; inner edge >= 9 mm off the frame seam (now 9.5 mm at the tightest; part A clamp 0 mm). Bottom and sides 301, box pleats laid flat, bartack the 2 mouth corners.	FP x2, B-03 x2	WF1	Nm 90/14 DPx17, medium ball SUK (knit mesh)	BN tex70	10	301 / LSd	1250	SK-5; FP corners (BT-S)	1.15 / 8.3
4.10	Gusset flat work: IMP-8 (R only) and hose ports	IMP-8 cup at 'IMP-8 cup top (R only)'. Hose slots (both gussets, z 700): B-09p binding (TPU, laser-cut) laid over the slot, perimeter + slot edge stitched.	IMP-8, B-09p x2, B-03 x2	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / LSd	1090		0.81 / 5.9
4.11	Frame-side box seams B-01 | gussets	R (720 mm): E5 pole-grip tab at z 640 and the B-26 shock-cord ends (15 mm in) inserted. L (720 mm): 2 HPL hard-point loops at z 420 / 640 inserted. Notch to notch every 50 mm; SSa2.	B-01, B-03 x2, E5, E5-cord, B-26, HPL x2	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / SSa-ins + SSa2	2881	SK-5; QC-08	1.16 / 8.4
4.12	Aft corner seams blade U | gussets (closes the tube)	R and L (715 / 715 mm): 2 HPL loops in the L seam at the hard-point notches; 3 K-W keeper loops in each seam at W1-W3 heights (ASSUMPTION KW-PLACE: W heights z 80 / 250 / 420 from the frame lash points; not drawn). SSa2.	B-02u, B-03 x2, HPL x2, K-W x6	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / SSa-ins + SSa2	2861	SK-5; QC-08	1.66 / 11.9
4.13 V	US floor channels B-16 on F-02	Turn both ends of each B-16 15 mm (left OPEN: the 25 mm under-strap passes through), lay it across F-02's outer face on the drawn 'B-16 channel' lines (stations [48.0, 104.0] mm), topstitch both long edges, bartack the 4 corners.	B-16 x2, F-02	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / LSd	760		1.01 / 7.2
4.14	Floor B-05 (F-02 first)	Sew F-02 to the outside of B-05. Then the floor ring (1107 mm): B-01 edge first with the HE tails and the B-18 lower edges caught in it, then gussets, then the blade kick strip; quarter notches matched, needle down at the 4 corners. SSa2. US channels: on F-02 (op above).	B-05, F-02, B-01, B-03 x2, B-02u	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / SSa2	2874	QC-08	1.89 / 13.6
5.01	Collar tube	B-06f (ULTRA) to B-06 both sides (990 mm); I2T hang tab at the B-06f drill mark.	B-06f, B-06, I2T	WF1	Nm 90/14 DPx17, round point R	BN tex70	10	301 / SSa2	1979		0.69 / 5.0
5.02	Roll-top hem with HDPE strips and clip straps	Close the hem back seam (70 mm). Sew the hem to the collar top ring (1050 mm), fold over the two B-07s HDPE strips and stitch closed (EFb). RT clip straps (510 mm) at each roll end: 60 mm sewn to the roll end, 30 mm round the side-release buckle bar, box-X.	B-07, B-07s x2, B-06, RT x2, 2 SR buckles	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / SSa + EFb + BOX	2484	SK-2	1.37 / 9.9
5.03	Collar ring to the body	Collar to the body tops (1050 mm), matched at every panel-corner V-notch. The seam closes the top end of each B-27 port; bartack the port top over it.	collar, body	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / SSa2	2100	QC-08, QC-10	1.33 / 9.6
6.01	Lid: 4 crown darts sewn + pressed	Fold each B-29a corner dart on its centre, sew leg to leg from the edge to the apex drill mark, tie off (no backtack at the apex), press the dart toward the aft edge. (rows from patterns_corrie_f_rev3.lid_hours)	B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / DART		SK-6	1.00 / 7.2
6.02	Lid: strap channels topstitched to the crown, both edges (2 x 2 runs)	Lay each B-29g on its 'strap channel CL' line (x +/-52), edges turned 8 mm, topstitch both long edges; channels are 30 mm wide for the 25 mm B-28 strap (drawn so). (rows from patterns_corrie_f_rev3.lid_hours)	B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / LSd	1062	SK-6	0.64 / 4.6
6.03	Lid: bartacks at the 4 channel ends	BT-M bartack across each channel end (tex 90, 304), 3 mm in from the turned end. (rows from patterns_corrie_f_rev3.lid_hours)	B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BT		SK-6	0.40 / 2.9

Seq	Operation	How (beginner)	Pieces	Machine	Needle	Thread	SPI	Stitch / seam	mm	Ref	h / SAM min
6.04	Lid: camo caps: short ends turned and topstitched (2 x 2)	Turn the short ends of each B-29f 10 mm and topstitch the cap on B-29e at its placement; long edges go into the ring seams. (rows from patterns_corrie_f_rev3.lid_hours)	B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / LSd	60	SK-6	0.05 / 0.4
6.05	Lid: #5 coil zip: tape baste, stops, slider	Baste the #5 zip between B-29d (upper) and B-29e (lower) with 6 mm tape; zip stops at the drill marks; slider on before closing. (rows from patterns_corrie_f_rev3.lid_hours)	B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / ZIP		SK-6	0.50 / 3.6
6.06	Lid: #5 coil zip sewn, both tapes	ZIP seam: each tape at 5 mm, turn, edge-stitch 3 mm. (rows from patterns_corrie_f_rev3.lid_hours)	B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / ZIP	940	SK-6	0.47 / 3.4
6.07	Lid: zip round the two aft corners	Ease the zip round each aft corner; clip the tape 3 mm at the corner tangent notches. (rows from patterns_corrie_f_rev3.lid_hours)	B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / ZIP		SK-6	0.80 / 5.8
6.08	Lid: wall ring closed: two short seams	Join the front wall run B-29c to the aft strips at both ends (30 mm seams), SSa2. (rows from patterns_corrie_f_rev3.lid_hours)	B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / SSa2	120	SK-6	0.04 / 0.3
6.09	Lid: crown | wall perimeter (curved, notch match, bind)	Crown to wall ring notch to notch, darts matched to the corner notches; clip the crown allowance every 15 mm on the curves; crown on top. (rows from patterns_corrie_f_rev3.lid_hours)	B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / SSa2	1978	SK-6	1.78 / 12.8
6.10	Lid: crown | wall: closed ring handling	Sew the ring in one pass, needle down at each notch; open the zip first. (rows from patterns_corrie_f_rev3.lid_hours)	B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / SSa2		SK-6	0.50 / 3.6
6.11	Lid: wall | base perimeter (curved, notch match, bind), turned through the zip	Base B-29b to the wall ring, notch to notch, inside out; then turn the lid right side out through the open zip. (rows from patterns_corrie_f_rev3.lid_hours)	B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g	WF1	Nm 90/14 DPx17, round point R	BN tex70	10	301 / ZIP	1978	SK-6	1.78 / 12.8
6.12	Lid: wall | base: closed ring handling	As the crown ring: one pass, needle down at each notch. (rows from patterns_corrie_f_rev3.lid_hours)	B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g	WF1	Nm 90/14 DPx17, round point R	BN tex70	10	301 / SSa2		SK-6	0.50 / 3.6
7.01	B-28: B-28 crown strap, kit M x2	straight strap, sealed ends; threaded at assembly (op 7.end).	B-28 x2	bench	-	-	-	- / -		SK-1, SK-7	0.00 / 0.0
7.02	B-28t: B-28 lock tab (front on the hinge band, rear through the B-25 toggle, B-25 second fold) x6	60 mm (2.4 in). Folds: fold round the lock bar / toggle. Box-X: 21 x 20 at the drawn marks.	B-28t x6	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	840	SK-1, SK-7; QC-07; B-28t	0.60 / 4.3
7.03	K-28: B-28 tail keeper (25 mm elastic loop) x4	Cut 90 mm, lap 15 mm into a loop round its strap, zigzag the lap (3 passes).	K-28 x4	ZZ	Nm 110/18 DPx17, round point R	BN tex90	8	304 / BT (zigzag lap)			0.40 / 2.9
7.04	W1: haul wrap W1 (cut for carry) x1	750 mm (29.5 in). Folds: G-hook fold. Box-X: 21 x 25 at the drawn marks.	W1 x1	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	157	SK-1, SK-7; QC-07; W1	0.10 / 0.7
7.05	W2: haul wrap W2 (cut for carry) x1	822 mm (32.4 in). Folds: G-hook fold. Box-X: 21 x 25 at the drawn marks.	W2 x1	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	157	SK-1, SK-7; QC-07; W2	0.10 / 0.7
7.06	W3: haul wrap W3, kit M x1	987 mm (38.8 in). Folds: G-hook fold. Box-X / bartack per the load table where it is caught in a seam.	W3 x1	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	157	SK-1, SK-7; QC-07; W3	0.10 / 0.7
7.07	WX: W haul extender (meat-trip kit) x3	740 mm (29.1 in). Folds: G-hook fold; sewn loop fold (takes the W's own G-hook). Box-X: 21 x 25, 21 x 25 at the drawn marks.	WX x3	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	944	SK-1, SK-7; QC-07; WX	0.60 / 4.3
7.08	K-W: W gore keeper loop (6, in the gore seams) x6	80 mm (3.1 in). Folds: fold, raw ends in the seam. Box-X / bartack per the load table where it is caught in a seam.	K-W x6	BT	Nm 110/18 DPx17, round point R	BN tex90	-	304 / BT		SK-1, SK-7	0.60 / 4.3

Seq	Operation	How (beginner)	Pieces	Machine	Needle	Thread	SPI	Stitch / seam	mm	Ref	h / SAM min
7.09	RT: roll-top side clip strap x2	510 mm (20.1 in). Folds: sewn to the roll end (60); round the buckle bar (30). Box-X / bartack per the load table where it is caught in a seam.	RT x2	BT	Nm 110/18 DPx17, round point R	BN tex90	-	304 / BT		SK-1, SK-7	0.20 / 1.4
7.10	K-RT: roll clip tail keeper (25 mm elastic) x2	Cut 90 mm, lap 15 mm into a loop round its strap, zigzag the lap (3 passes).	K-RT x2	ZZ	Nm 110/18 DPx17, round point R	BN tex90	8	304 / BT (zigzag lap)			0.20 / 1.4
7.11	IM1: internal meat strap IM1 x1	990 mm (39.0 in). Folds: left end loop (girth-hitched on the eye). Box-X / bartack per the load table where it is caught in a seam.	IM1 x1	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	157	SK-1, SK-7; QC-07; IM1	0.10 / 0.7
7.12	IM2: internal meat strap IM2 x1	1023 mm (40.3 in). Folds: left end loop (girth-hitched on the eye). Box-X / bartack per the load table where it is caught in a seam.	IM2 x1	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	157	SK-1, SK-7; QC-07; IM2	0.10 / 0.7
7.13	IM3: internal meat strap IM3 x1	1083 mm (42.6 in). Folds: left end loop (girth-hitched on the eye). Box-X / bartack per the load table where it is caught in a seam.	IM3 x1	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	157	SK-1, SK-7; QC-07; IM3	0.10 / 0.7
7.14	K-IM: IM tail keeper (25 mm elastic) x3	Cut 90 mm, lap 15 mm into a loop round its strap, zigzag the lap (3 passes).	K-IM x3	ZZ	Nm 110/18 DPx17, round point R	BN tex90	8	304 / BT (zigzag lap)			0.30 / 2.2
7.15	IME: IM fairlead eye (38 mm) x6	200 mm (7.9 in). Folds: fold (eye). Box-X: 34 x 30 at the drawn marks.	IME x6	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	1312	SK-1, SK-7; QC-07; IME	0.60 / 4.3
7.16 V	US: under-strap US1 / US2 x2	straight strap, sealed ends; threaded at assembly (op 7.end).	US x2	bench	-	-	-	- / -		SK-1, SK-7; QC-07; US	0.00 / 0.0
7.17	B-24: float stand-off link (38 mm, meat-trip) x2	435 mm (17.1 in). Folds: T3 bag hook loop 120; loop 170; loop 220. Box-X / bartack per the load table where it is caught in a seam.	B-24 x2	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	315	SK-1, SK-7; QC-07; B-24	0.20 / 1.4
7.18	HE: hinge eye (38 mm) x2	150 mm (5.9 in). Folds: fold back (eye); fold. Box-X: 34 x 34 at the drawn marks.	HE x2	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	464	SK-1, SK-7	0.20 / 1.4
7.19	T3: IM3 bag-hook tongue (38 mm) x2	160 mm (6.3 in). Folds: fold round the hook bed. Box-X: 34 x 30, 34 x 34 at the drawn marks.	T3 x2	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	902	SK-1, SK-7; QC-07; T3	0.40 / 2.9
7.20	B-25: guy bridle on the crown top bar x1	412 mm (16.2 in). Folds: bar sleeve R; bar sleeve L. Box-X / bartack per the load table where it is caught in a seam. FRAME COORDINATION: the bar sleeves close round the crown top bar BEFORE the crown corners are bonded (H1 build); sew the toggle tabs + second folds here, the sleeves at frame assembly.	B-25 x1	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	157	SK-1, SK-7; QC-07; B-25	0.10 / 0.7
7.21	TG: port guy (B-25 toggle -> B-27 port -> B-22 inside) x2	500 mm (19.7 in). Folds: -. Box-X: 21 x 25 at the drawn marks.	TG x2	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	315	SK-1, SK-7; QC-07; TG	0.20 / 1.4
7.22	LLT: guy ladder-lock tab on B-22 x2	70 mm (2.8 in). Folds: fold round the lock bar. Box-X: 21 x 25 at the drawn marks.	LLT x2	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	315	SK-1, SK-7	0.20 / 1.4
7.23	K-G: guy tail keeper (25 mm elastic) x2	Cut 90 mm, lap 15 mm into a loop round its strap, zigzag the lap (3 passes).	K-G x2	ZZ	Nm 110/18 DPx17, round point R	BN tex90	8	304 / BT (zigzag lap)			0.20 / 1.4
7.24	GBL: guy bar loop (POM toggle) x2	170 mm (6.7 in). Folds: fold round the bar. Box-X: 21 x 25 at the drawn marks.	GBL x2	BT / PS (prototype WF2 + J1)	Nm 140/22 DPx17, round point R	BN tex135	7	301 box-X path / BOX	315	SK-1, SK-7; QC-07; GBL	0.20 / 1.4
7.25	HPL: scope-pod hard-point loop x4	80 mm (3.1 in). Folds: fold, raw ends in the seam. Box-X / bartack per the load table where it is caught in a seam.	HPL x4	BT	Nm 110/18 DPx17, round point R	BN tex90	-	304 / BT		SK-1, SK-7	0.40 / 2.9

Seq	Operation	How (beginner)	Pieces	Machine	Needle	Thread	SPI	Stitch / seam	mm	Ref	h / SAM min
7.26	E5: pole grip tab (cord lock) x1	60 mm (2.4 in). Folds: fold. Box-X / bartack per the load table where it is caught in a seam.	E5 x1	BT	Nm 110/18 DPx17, round point R	BN tex90	-	304 / BT		SK-1, SK-7	0.10 / 0.7
7.27	I2T: IMP-2 hang tab (B-06f) x1	100 mm (3.9 in). Folds: fold. Box-X / bartack per the load table where it is caught in a seam.	I2T x1	BT	Nm 110/18 DPx17, round point R	BN tex90	-	304 / BT		SK-1, SK-7	0.10 / 0.7
7.28	I2K: IMP-2 bottom keeper tab (B-01) x2	60 mm (2.4 in). Folds: fold. Box-X / bartack per the load table where it is caught in a seam.	I2K x2	BT	Nm 110/18 DPx17, round point R	BN tex90	-	304 / BT		SK-1, SK-7	0.20 / 1.4
7.29	BRL: bow riser loop (through B-11) x2	100 mm (3.9 in). Folds: fold. Box-X / bartack per the load table where it is caught in a seam.	BRL x2	BT	Nm 110/18 DPx17, round point R	BN tex90	-	304 / BT		SK-1, SK-7	0.20 / 1.4
7.30	BRS: bow riser strap + G-hook x1	300 mm (11.8 in). Folds: G-hook fold. Box-X / bartack per the load table where it is caught in a seam.	BRS x1	BT	Nm 110/18 DPx17, round point R	BN tex90	-	304 / BT		SK-1, SK-7	0.10 / 0.7
7.31	F-EL: fly hem elastic, 6 mm (spine top + bottom, sides bottom + frame) x1	Cut 2436 mm, lap 15 mm into a loop round its strap, zigzag the lap (3 passes).	F-EL x1	ZZ	Nm 110/18 DPx17, round point R	BN tex90	8	304 / BT (zigzag lap)			0.10 / 0.7
7.32	Fit the hardware	IM1/IM2: G-hook right end, cam threaded; IM3: wiregate carabiner on the free end; T3 loops: wiregate carabiner + TPU sleeve; W1-W3: rated G-hook end, heavy ladder lock on the gusset side; guys: toggle in the B-25 tab, through the B-27 port, heavy ladder lock on LLT, tri-glide keeper; B-28: toggle rear, ladder lock both ends, 2 elastic keepers; RT: side-release buckles; E5: cord lock.	hardware	bench	-	-	-	- / -		QC-11	1.00 / 7.2
8.01	Scope pod A1	Box pleats in A1-b, leading-edge bar A1-d on the sleeve mouth (EFa), sleeve face to back panel A1-a round the perimeter; 4 hooks / loops to the HPL hard points.	A1-a, A1-b, A1-d	WF1	Nm 110/18 DPx17, round point R	BN tex90	8	301 / SSa2 + EFa	3770		1.63 / 11.8
8.02	Meat liner MS (SWAP slab)	Tube MS-a to frame face MS-b and floor MS-c; HDPE MS-s in the roll pocket. Silpoly: clips, no pins; 10 SPI tex 70.	MS-a, MS-b, MS-c, MS-s x2	WF1	Nm 90/14 DPx17, round point R	BN tex70	10	301 / SSa2	5280		2.08 / 15.0
9.01 V	F-01 fly: panel seams	Join the fly panels (F-01a spine R | F-01b side 1271 mm; F-01a spine L | F-01b side 1271 mm). Silpoly: weights and clips, no pins; 301 then flat-fell; seam-seal the felled seams (piece note: silpoly: seams 301 then flat-felled + seam-sealed; 20 mm elastic channel hems top + bottom)	F-01a, F-01b	WF1	Nm 90/14 DPx17, round point R	BN tex70	10	301 / SSa then LSc flat-fell (ISO 4916 class 2)	7628		3.05 / 22.0
9.02 V	F-01 fly: elastic channel hems	Turn the hems 20 mm on the drawn hem lines (3249 mm of hem); thread 2436 mm of 6 mm elastic (F-EL), bar each end.	F-01a, F-01b, F-EL	WF1	Nm 90/14 DPx17, round point R	BN tex70	10	301 / EFb channel	3248		2.35 / 16.9
10.01	Trim, seal, turn	Trim thread ends to 3 mm, heat-seal webbing ends that frayed, turn the bag right side out through the door.		bench	-	-	-	- / -			0.50 / 3.6
10.02	Final inspection and weigh	QC sheet sections F and M; weigh each kit against the ledger.		bench	-	-	-	- / -		QC-12 to QC-20	0.50 / 3.6
10.03	Pack	README.md, Packaging section.		bench	-	-	-	- / -			0.20 / 1.4

7. Load points: bartack and box-X specification

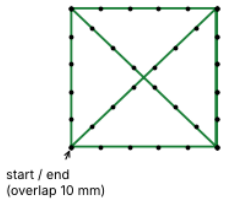
Load = factored design load from the record named (x 3.0 on the 1 g simulation per HA-LS-1 rev C). Capacity by attach_system rules. Proof = the test that releases the point (register).

Point	Part	Where	Pattern	Stitches	Machine	Thread / needle	Load N (lbf)	Source	Capacity N	Governs	SF	Proof
port guy, toggle end	TG	TG toggle loop (B-25 tab side)	box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm	48	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	288.0 (65.0)	corrie_f_body.md 13.5: 96.1 N x 3.0	3186.0	box-X	11.06	LL-H (366 N, toggle-loop pull, loadpath A3)
port guy, bag end	LLT + B-22 + B-02h	inside the hinge band at x +/-130.3	box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm	48	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	288.0 (65.0)	corrie_f_body.md 13.5: 96.1 N x 3.0	1024.0	B-22 perimeter 50 mm (primary body)	3.56	LL-H; PT-5 (lock + keeper); alternate body fabric: B-22 seam 1408 N
port guy, B-27 slit ends	B-01	both ends of each slit, z 595 / 745	bartack 10 x 2 mm, 304	28	BT LK-1900BN	bonded_nylon_tex90 / Nm 110/18 DPx17, round point R	- (-)	no rated load (lap only); port mock-up: 100 N guy pull through the lap		-		port mock-up (13.9)
crown strap tabs (6)	B-28t	front: hinge band (outside); rear: through the B-25 toggle; B-25 second fold	box-X 21 x 20 mm (0.83 x 0.79 in), path 140 mm	43	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	288.0 (65.0)	corrie_f_body.md 15.2: 288 N envelope (port guy) if a crown strap ever holds the top	2836.0	box-X	9.85	LL-H (heavy lock slip 366 N covers 288 N; BL-031 dropped the 20 mm gate)
crown strap front tab on the band	B-02h	B-28 front tab box on the hinge band	box-X 25 x 25 mm (0.98 x 0.98 in), path 171 mm	52	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	288.0 (65.0)	as above	3458.0	box-X	12.01	LL-H
B-25 bridle tab folds (4)	B-25	toggle tab + second fold, each side (not drawn on the pattern)	box-X 21 x 20 mm (0.83 x 0.79 in), path 140 mm	43	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	288.0 (65.0)	corrie_f_body.md 13.5 (guy) / 15.2 (crown strap)	2836.0	box-X	9.85	LL-H toggle-loop pull 366 N; ASSUMPTION B25-BOX: box-X 21 x 20 closes each fold; pattern shows folds only
guy bar loop	GBL	round the crown top bar	box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm	48	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	244.0 (55.0)	loadpath_verification A3: 244 N	3186.0	box-X	13.06	LL-H
IM3 bag-hook tongue	T3	loop 34x30	box-X 34 x 30 mm (1.34 x 1.18 in), path 219 mm	65	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	1702.0 (383.0)	loadpath_verification A1: IM3 1702 N factored (guys-off misuse)	4000.0	carabiner (PT-1 basis: 0.5 x 8 kN minor axis)	2.35	PT-1 2553 N carabiner; PT-2 2553 N
IM3 bag-hook tongue	T3	box-X to B-17 34x34	box-X 34 x 34 mm (1.34 x 1.34 in), path 232 mm	68	PS AMS-210EN (or WF2 + J1)	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	1702.0 (383.0)	loadpath_verification A1: IM3 1702 N factored (guys-off misuse)	4000.0	carabiner (PT-1 basis: 0.5 x 8 kN minor axis)	2.35	PT-1 2553 N carabiner; PT-2 2553 N
B-17 IM3 tab perimeter (double row)	B-17	OUTSIDE B-01 at x +/-115, z 385-455	double-row perimeter 3 / 9 mm, 301		WF2	bonded_nylon_tex90 / Nm 110/18 DPx17, round point R	1702.0 (383.0)	loadpath_verification A1	3277.0	perimeter 80 mm loaded, 2 rows	1.93	PT-2 2553 N on the 70 mm tab; perimeter 3 / 9 mm, 8 SPI tex 90; SF from the 90 x 110 basis, PT-2 settles the 70 mm tab
IM fairlead eyes (6)	IME	frame side, captive in the printed fairleads	box-X 34 x 30 mm (1.34 x 1.18 in), path 219 mm	65	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	1702.0 (383.0)	IM3 eye carries the bag hook (A1 1702 N); IM1/IM2 605 N (S1)	4430.0	box-X	2.6	MT-1 analogue at 38 mm; IM3-EP 2553 N
hinge eye	HE + B-18 + B-01	x +/-115, eye top z 60	box-X 34 x 34 mm (1.34 x 1.34 in), path 232 mm	68	PS AMS-210EN (or WF2 + J1)	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	795.0 (179.0)	loadpath_verification A5: 795 N	2355.0	B-18 seam (80 mm, B-05 strip)	2.96	PT-3 1530 N
IM1 end loop	IM1	left end loop (girth hitch on the eye); not drawn as a box on the pattern	box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm	48	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	605.0 (136.0)	loadpath_verification S1: 605 N	3186.0	box-X	5.27	MT-1 908 N; ASSUMPTION IM-BOX: box-X 21 x 25 closes the 60 mm loop (MT-1 specimen)

Point	Part	Where	Pattern	Stitches	Machine	Thread / needle	Load N (lbf)	Source	Capacity N	Governs	SF	Proof
IM2 end loop	IM2	left end loop (girth hitch on the eye); not drawn as a box on the pattern	box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm	48	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	605.0 (136.0)	loadpath_verification S1: 605 N	3186.0	box-X	5.27	MT-1 908 N; ASSUMPTION IM-BOX: box-X 21 x 25 closes the 60 mm loop (MT-1 specimen)
IM3 end loop	IM3	left end loop (girth hitch on the eye); not drawn as a box on the pattern	box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm	48	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	605.0 (136.0)	loadpath_verification S1: 605 N	3186.0	box-X	5.27	MT-1 908 N; ASSUMPTION IM-BOX: box-X 21 x 25 closes the 60 mm loop (MT-1 specimen)
US1/US2 sewn eye	US	eye on the left pin	box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm	48	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	720.0 (162.0)	loadpath_verification S5-S7: 720 N	3186.0	box-X	4.43	MT-1 1080 N; ASSUMPTION US-BOX: box-X 21 x 25; pattern has no marks
W1 G-hook end	W1	G-hook fold 2 x 30 mm	box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm	48	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	225.0 (51.0)	loadpath_verification S9: 225 N	3186.0	box-X	14.16	LL-H 366 N; W-PULL 338 N
W2 G-hook end	W2	G-hook fold 2 x 30 mm	box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm	48	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	225.0 (51.0)	loadpath_verification S9: 225 N	3186.0	box-X	14.16	LL-H 366 N; W-PULL 338 N
W3 G-hook end	W3	G-hook fold 2 x 30 mm	box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm	48	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	225.0 (51.0)	loadpath_verification S9: 225 N	3186.0	box-X	14.16	LL-H 366 N; W-PULL 338 N; ASSUMPTION W3-BOX (pattern draws the fold only)
W haul extender	WX	box-X hook end	box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm	48	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	225.0 (51.0)	ASSUMPTION WX-LOAD: same float-clamp load as S9 (225 N)	3186.0	box-X	14.16	LL-H; one extender pulled to 338 N
W haul extender	WX	box-X loop	box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm	48	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	225.0 (51.0)	ASSUMPTION WX-LOAD: same float-clamp load as S9 (225 N)	3186.0	box-X	14.16	LL-H; one extender pulled to 338 N
B-24 float link: hook end	B-24	38 mm, hook fold (pattern: folds only)	box-X 34 x 34 mm (1.34 x 1.34 in), path 232 mm	68	PS AMS-210EN (or WF2 + J1)	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	941.0 (212.0)	loadpath_verification S12 / RD-02: 941 N misuse	4000.0	carabiner	4.25	PT-6 1410 N; attach_system B-24: box-X 38 x 38 (hook) + 3 x 38 x 20 (loops); drawn here 34 wide (2 mm edge)
B-24 float link: 3 loops	B-24	loop beds 120 / 170 / 220 mm	box-X 34 x 20 mm (1.34 x 0.79 in), path 187 mm	56	BT LK-1900BN	bonded_nylon_tex135 / Nm 140/22 DPx17, round point R	941.0 (212.0)	as above	3786.0	box-X	4.02	PT-6 1410 N

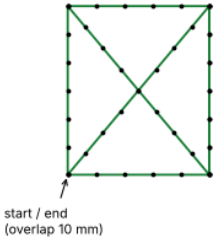
SK-1 Load-point stitch patterns (true proportions; dots = 7 SPI stitch positions)

box-X 21 × 20 mm (0.83 × 0.79 in)



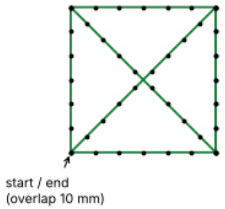
43 stitches incl. 4 lock, 7 SPI, tex 135, Nm 140 R
used on: B-25, B-28t

box-X 21 × 25 mm (0.83 × 0.98 in)



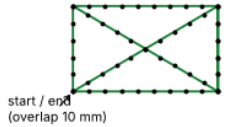
48 stitches incl. 4 lock, 7 SPI, tex 135, Nm 140 R
used on: GBL, IM1, IM2, IM3, LLT + B-22 + B-02h, TG, US, W1, W2, W3, WX

box-X 25 × 25 mm (0.98 × 0.98 in)



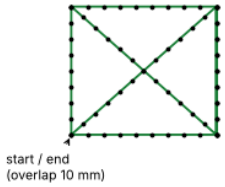
52 stitches incl. 4 lock, 7 SPI, tex 135, Nm 140 R
used on: B-02h

box-X 34 × 20 mm (1.34 × 0.79 in)



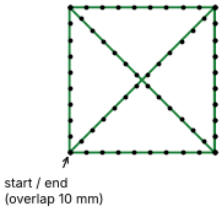
56 stitches incl. 4 lock, 7 SPI, tex 135, Nm 14
used on: B-24

box-X 34 × 30 mm (1.34 × 1.18 in)



65 stitches incl. 4 lock, 7 SPI, tex 135, Nm 140 R
used on: IME, T3

box-X 34 × 34 mm (1.34 × 1.34 in)



68 stitches incl. 4 lock, 7 SPI, tex 135, Nm 140 R
used on: B-24, HE + B-18 + B-01, T3

BT-S bartack 10 × 2 mm, 28 st (304)



LK-1900 standard tack, tex 90, Nm 110 R
B-27 slit ends, FP mouth corners, I2K

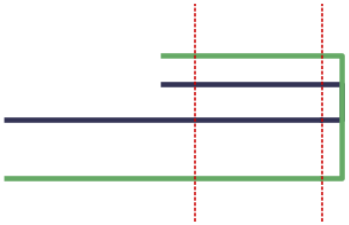
BT-M bartack 16 × 2 mm, 42 st (30



LK-1900 standard tack, tex 90, Nm 110 R
B-27 slit ends, FP mouth corners, I2K

SK-2 Seam constructions (cross-sections, not to scale; red dashed = stitch row)

SSa2 body seam: 12 mm, then allowances turned and caught at 6 mm (2 rows)



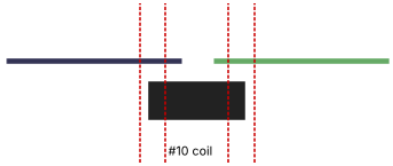
LSd Hypalon patch: double row 3 / 9 mm from the edge, 8 SPI tex 90



EFb roll-top hem round the HDPE strip



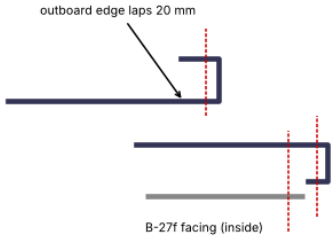
ZIP: tape sewn at 5 mm, turned, edge-stitched 3 mm (door, lid, pockets)



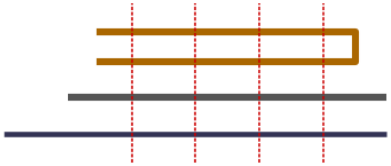
SSa-ins: loop / tab caught in the seam, backtack each side (HPL, K-W, E5)



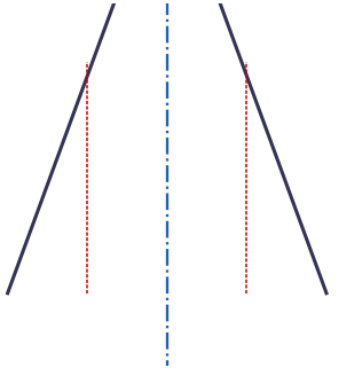
B-27 guy port: both edges turned 6 mm, facing on the inboard edge, bartack ends



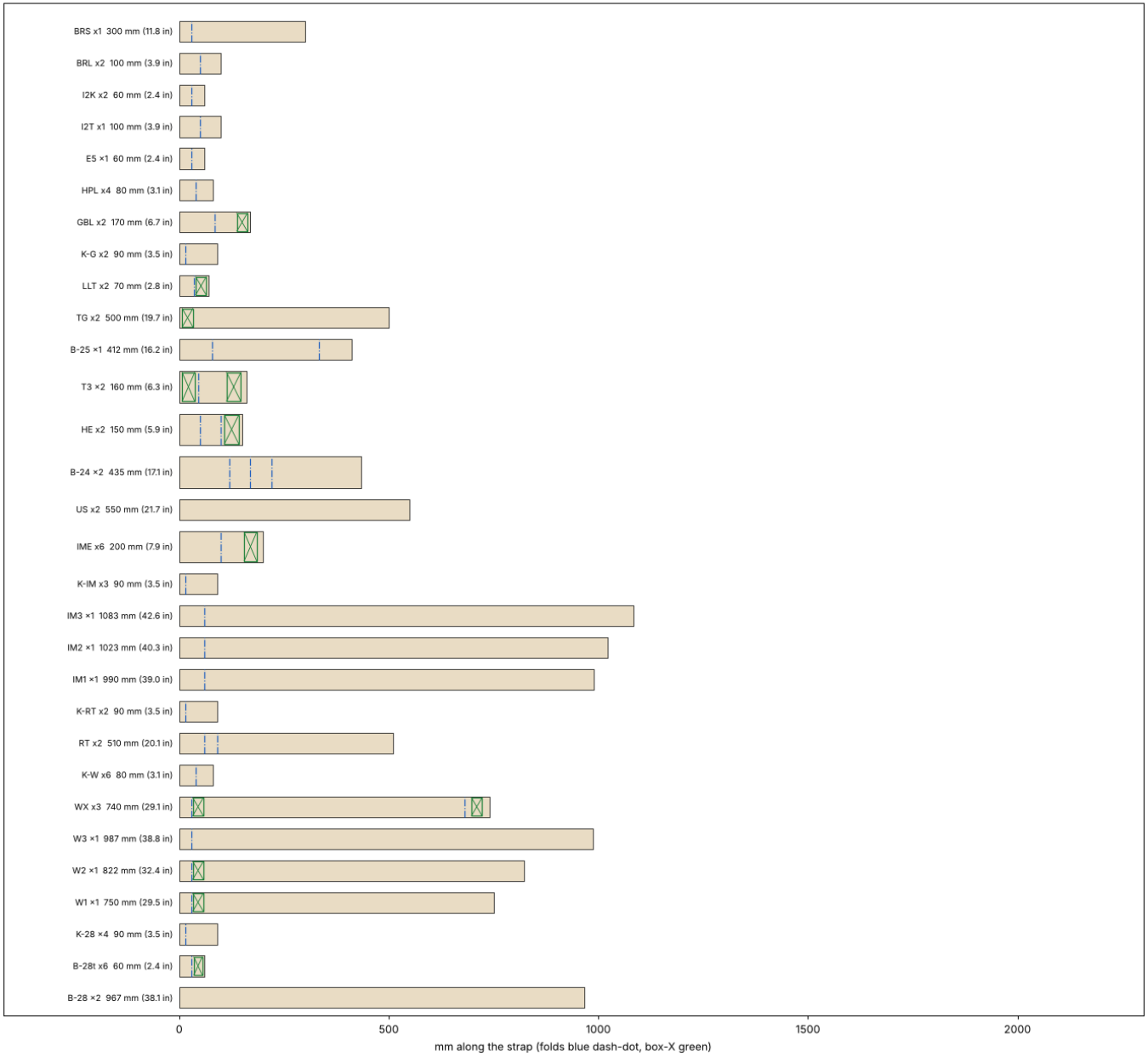
Load stack (e.g. LLT + B-22 + B-02h): webbing fold / Hypalon / panel, box-X through all



DART (B-29a, 4 corners): fold on the centre, sew leg to leg, tie off at the apex



SK-7 Webbing parts as cut (crown kit M); hot-knife length = strap length



8. QC checkpoints

ID	Stage	Check	Method	Pass	Frequency
QC-01	Print	Test square	ruler on 00_PRINT_FIRST and on one page per print run	100.0 +/- 1.0 mm both ways	every print run
QC-02	Incoming	Fabric weight and width	cut 100 x 100 mm, weigh 0.01 g; tape the roll width	gsm within +/- 5 % of the record; width >= usable_width_mm	every roll
QC-03	Cutting	Piece size and cut-outs	lay the pattern on the cut piece	+/- 1 mm on pieces < 300 mm, +/- 2 mm on larger; windows / slots / slits present and clean	first piece of each lay + 1 in 10
QC-04	Cutting	Notches, grain, webbing length	visual; tape	notch 6 mm (never past the seam line); grain within 2 deg of the arrow; webbing +/- 2 mm, ends sealed	first of each + 1 in 10
QC-05	Sewing	Stitch density	count stitches over 25.4 mm	body 8, light 10, box-X 7 SPI; +/- 1	start of each shift / each op set-up
QC-06	Sewing	Tension balance	cut a swatch seam open	lock mid-stack, no loops on either face	each set-up
QC-07	Load points	Box-X and bartacks	visual against SK-1; count	full rectangle + both diagonals, start / end overlap >= 10 mm, stitch count >= load table, no skipped stitch, needle hole not torn	100 %
QC-08	Assembly	Seam match	notches at both ends of every 50 mm	notch offset <= 2 mm; seam end mismatch <= 3 mm; allowance 12 +/- 1 mm	100 %
QC-09	Assembly	Door zip	run both sliders 10 x end to end; look at the corners	no snag, no wave > 3 mm at the R 55 corners, tape not caught, stops at the hinge band	100 %
QC-10	Assembly	B-27 ports	pull the guy through at 45 deg; look inside	20 mm lap closes flat; both ends bartacked; facing follows the crease at z 631	100 %
QC-11	Hardware	Hardware fitted	count against the BOM; threading as SK-7	every lock threaded the right way (pull test by hand), carabiners gate-in, cord lock holds	100 %
QC-12	Final	Width	tape across the widest point, bag packed full	<= 394 mm (15.5 in) (owner limit)	first article + 1 in 10
QC-13	Final	Volume	fill with 10 mm foam balls / litre jug method, collar rolled 3 times	85 L +/- 3 L (5200 cu in) (owner limit)	first article
QC-14	Final	Mass, bag base	1 g scale	within +/- 3 % of the ledger base bag (1065 g on the ledger basis; primary fabrics are heavier, see costing.md)	every bag
QC-15	Final	No hook-and-loop, coil zips only	visual	zero hook-and-loop anywhere (owner: quiet bag)	100 %
QC-16	Final	Camo use	visual	camo only on B-02u, SH-07, SH-01b, FP-b, B-29f, A1-d (accents)	100 %
QC-17	Final	Thread ends, seals	visual	ends <= 3 mm, no open webbing ends, no burnt fabric	100 %
QC-18	Final	Labels	visual	ID label with rev 3 + date, size / crown kit, load-limit card (53 kg frame meat, RD-02)	100 %
QC-19	Final	Kits complete	count	base / as-hunted (lid, B-28 per kit, scope pod) / meat-trip (WX, B-24, liner) bags each complete	100 %
QC-20	Final	Fit on the frame	hook to a frame or the fit jig: hinge pins, IM eyes through B-12 ports, guys through B-27	every eye reaches its port without strain; door opens fully	first article + 1 in 25

Physical tests that gate release

ID	Test	Gates	Steps	Pass / fail	Source
MT-1	25 mm box-X bartack proof at the BL-010 per-row loads	every 25 mm box-X on IM1-IM3 and US1/US2	sew 10 loops exactly like the pack; loop over the 5 mm pin; 5 loops to 908 N (92.6 kg) and 5 to 1080 N (110.1 kg); hold 1 min each	5 of 5 hold 908 N (IM) and 5 of 5 hold 1080 N (US) for 1 min, no broken stitch	test_register.csv
MT-2	meat-strap hook proof per strap (BL-010)	IM free-end hooks	as PT-1: G-hooks to 645 N (65.7 kg, covers IM1 410 N), carabiners to 910 N (92.8 kg); hold 1 min	5 of 5 G-hooks hold 645 N and 5 of 5 carabiners hold 910 N for 1 min, gates close	test_register.csv
MT-3	25 mm cam buckle slip at the BL-010 per-row loads	IM / US cams	thread the cam as in the pack; paint-pen mark at the cam mouth; pull to 908 N (92.6 kg), hold 1 min, measure; then to 1080 N (110.1 kg), hold 1 min, measure	mark moves < 3 mm at 908 N and at 1080 N, 5 of 5	test_register.csv
PT-1	IM3 bag carabiner on the sewn T3 tongue, per-row proof (BL-010)	IM3 carabiner on T3	carabiner over the 5 mm steel pin; sewn T3 tongue on the carabiner; pull with the come-along to 2553 N (260.2 kg on the scale); hold 60 s; release; check the gate closes and look for set	5 of 5 hold 2553 N for 60 s, gate closes after, no visible set	test_register.csv
PT-2	T3 tongue on the 70 mm B-17 tab on a B-01 swatch (BL-010 load)	T3 on the 70 mm B-17 tab	clamp the B-01 swatch 120 mm wide between bar clamps; tongue loop on the pin; pull to 2553 N (260.2 kg); hold 60 s	5 of 5 hold 2553 N, no stitch break, no laminate tear	test_register.csv
PT-3	HE eye on B-18 + B-01 + B-05 seam, built without binding	HE eye on B-18 (no binding)	5 mm pin through the eye; clamp the swatch; pull to 1530 N; hold 60 s	5 of 5 hold 1530 N	test_register.csv
PT-5	guy ladder lock + tri-glide keeper slip	guy lock + tri-glide keeper	mark webbing; 10 cycles 0 -> 170 N -> 0 by hand on the come-along; measure the mark	slip < 5 mm over 10 cycles	test_register.csv
PT-6	B-24 float stand-off link	B-24 float link	link hook on the pin; 38 mm loop through the 220 mm loop; pull in line to 1410 N; hold 60 s	5 of 5 hold 1410 N, no box-X break, gate closes	test_register.csv
LL-H	heavy 25 mm ladder lock slip, 1 h hold (guys and W1-W3)	heavy 25 mm ladder lock slip: guys, W1-W3 and the B-28 crown straps (the B-28 slip gate since BL-031)	thread as in the pack; paint-pen mark at the lock; hang 37.3 kg (366 N, sand bags) 50 mm above the floor for 1 h, all 5 side by side; measure the mark	slip < 5 mm after 1 h, 5 of 5 (366 N covers the W 338 N proof)	test_register.csv
T9	loaded CG (+ 15 in, 30 kg meat check; CORRIE F rev 3: port guys, crown strap cinch, lid)	rev 3 items: port guys at the S kit, crown strap cinch and free length, lid mass and seat	68 kg fill and meat on IM1-IM3 at 15/19/23 in; hang from haul loop, then from a hip pin, plumb both lines; add 30 kg on IM1-IM3 at 15 in; REV 3 (BL-030): (a) S kit (15 in), top full, port guys fitted, lid on with 1 kg, meat 30 kg on IM1-IM3 + fill to 68 kg: plumb-line CG both lines, record against 181.3 mm; (b) crown strap cinch: spring scale (0-150 N) on each strap's tail, pull to 80 N, lock; read again after 30 min of walking and after a 3 g drop-on-the-base (pack set down from 0.5 m); (c) cinched crown strap free length bar-tab to ladder lock at the S kit full and the L kit 6-day; (d) weigh the sewn lid alone (0.1 g scale) and photograph it on the packed 6-day and full bag from the rear 3/4 and the side	bag CG <= 205 mm at z <= 590; meat <= 190; tall <= 185; rev 3: CG <= 190 mm (record vs 181.3); cinch after the walk >= 40 N or the lid still seated on the full top (report the value); free length <= cut - 210 mm (1138 - 150 tail - 60 loop = 928 mm); lid 45.9 g +/- 5 g (record); lid edge on the hinge-band seam, caps over the blades	test_register.csv
FS-O5	ULTRA 200X 50 mm strip tensile (C-5 / O-5)	alternate fabric gate: ULTRA 200X 50 mm strip (floor / frame side)	clamp each strip in the 50 mm bar grips, 200 mm gauge; pull slowly to break; record the peak	mean >= 1840 N/50 mm, minimum >= 1195 N/50 mm	test_register.csv
B-28 slip	B-28 crown strap on its heavy ladder lock: covered by LL-H at 366 N (BL-031 dropped the 20 mm gate; 25 mm strap on ladder_lock_25mm_heavy)	release of this package	B-28 crown strap on its heavy ladder lock: covered by LL-H at 366 N (BL-031 dropped the 20 mm gate; 25 mm strap on ladder_lock_25mm_heavy)	LL-H pass	corrie_f_body.md 15.6 / 16
PORT-1	B-27 port mock-up: hose from above 1 min with the guy pulled at 45 deg; 100 N guy pull through the lap; gloved lock use at the port	release of this package	B-27 port mock-up: hose from above 1 min with the guy pulled at 45 deg; 100 N guy pull through the lap; gloved lock use at the port	no water inside; lap stays closed at 100 N	corrie_f_body.md 13.9
GT-1	capstan pull: 5 kg on the guy over the packed bag's corner	release of this package	capstan pull: 5 kg on the guy over the packed bag's corner	record MU_WEB_FAB (0.20-0.40 band)	corrie_f_body.md 13.9
ALT-1	alternate set: owner sees an ULTRAGrid swatch	release of this package	alternate set: owner sees an ULTRAGrid swatch	owner OK in writing	part A README section 1
ALT-2	alternate set: PT-3 on 200X + 50 mm strip test	release of this package	alternate set: PT-3 on 200X + 50 mm strip test	>= 36.8 N/mm (1840 N / 50 mm mean, FS-O5)	part A README; register FS-O5

ID	Test	Gates	Steps	Pass / fail	Source
ALT-3	alternate set: weigh the 70D stock	release of this package	alternate set: weigh the 70D stock	gsm within +/- 5 % of nylon_ripstop_70d record	part A README
FRAY-1	wash-swatch fray test of the un-bound SSa2 seam (3 wash cycles, R-12)	release of this package	wash-swatch fray test of the un-bound SSa2 seam (3 wash cycles, R-12)	no allowance frayed past the 2nd row; else bind (ASSUMPTION NO-BIND)	RD-17

Defect list

Defect	Where	Cause	Prevent / fix	Class
Skipped stitches in a box-X	load points	needle too small / wrong point, flagging on Hypalon	Nm 140 R, tex 135, walking foot pressure; re-sew the whole box offset 1 mm	critical
Box-X short (stitch count under the table)	load points	early stop, wrong program	count against the load table; re-sew	critical
Hardware threaded backwards (ladder lock, cam)	straps	assembly	QC-11 hand pull; SK-7	critical
Load tab on the wrong face (B-17 out, B-18 / B-22 in)	B-01, B-02h	face mix-up	placement labels say INSIDE / OUTSIDE; check before sewing	critical
Needle-cut Hypalon / laminate (perforation line)	B-17, B-18, B-22, B-01	cutting point needle or SPI too high	R point only; 8 SPI max on Hypalon	critical
Swell puckered or twisted in its window	B-01	window not notch-matched	unpick, re-clip at the corners	major
B-27 port gapes (lap < 20 mm)	B-01	facing sewn on the wrong line	re-sew on the drawn stitch line	major
Door zip waves / snakes at the corners	door	zip stretched while sewing	basting tape, corner notches, ease into the corner	major
Seam allowance frayed past the 2nd row	body seams	straight-knife cut nylon	laser / hot-knife cut or bind (FRAY-1)	major
Notch mismatch > 2 mm / seam end mismatch > 3 mm	box seams, collar ring	pieces stretched	re-clip notch to notch, walking foot	major
Mesh pocket over-stretched	FP	mesh pulled while sewing	baste the pleats first, sew with the mesh on top	minor
Thread tension loops on the underside	any	tension / bobbin	QC-06 swatch at each set-up	minor
Burn marks / hard beads on webbing ends	webbing	hot knife too hot / slow	J8 stop, steady cut	minor
Camo piece cut against the print direction	B-02u, SH-*	lay error	ONE WAY lay rule	minor
Hook-and-loop or a non-coil zip fitted	any	substitution	BOM check (owner: quiet bag)	critical

9. Packaging

1. Bag: collar unrolled, door zipped, straps coiled inside the bag (keepers on), folded flat to about 450 x 350 mm.
2. Base kit inside the bag: IM1-IM3, US1/US2, W1-W3 fitted; guys fitted to B-22 with the toggle end coiled.
3. As-hunted kit bag (poly bag 1, labelled): B-29 lid with the B-28 straps for the ordered crown kit (S / M / L), scope pod.
4. Meat-trip kit bag (poly bag 2, labelled 'MEAT TRIP KIT'): 3 WX extenders with G-hooks, 2 B-24 links with carabiners, meat liner MS.
5. Card on top: load limits (53 kg / 116 lb frame meat in float mode, RD-02), guy rule (port guys fitted for every carry, RD-23), the strap tail rule, wash cold / line dry, no hook-and-loop.
6. Outer: recycled poly bag 600 x 800 mm, 10 bags per carton; carton label: rev 3, door D2, crown kit, fabric set (primary / alternate), date, lot.

10. Costing and hours

Step	Beginner h
0 Prep and cutting	16.13
1 Pockets	6.37
2 Small parts	3.97
3 Door D2 and hinge band	5.00
4 Main bag	13.54
5 Collar and roll top	3.40
6 Crown lid B-29 (as hunted)	8.46
7 Straps and webbing	8.00
8 Add-ons (scope pod as hunted, meat liner meat trip)	3.71
9 Rain fly F-01	5.40
10 Finish, inspect, pack	1.20
Total (incl. one-off set-up)	75.2

Scenario	SAM min	Labour $	CMT incl. overhead $	Materials $	Total $
us_contract_usd_min	512	434.90	543.63	258.78	802.41
asia_contract_usd_min	512	61.40	76.75	258.78	335.53

Lid on the drafted pattern 9.53 h (booked 7.76 h). Alternate set materials $258.62. Assumptions: SAM-RATIO 0.12; LABOUR {'us_contract_usd_min': 0.85, 'asia_contract_usd_min': 0.12}; OVERHEAD 0.25; THREAD-FACTOR 2.75; WEB_WASTE 5 %; prices with no record: PRICE_ASSUMED.

11. Release delta register: rows this release resolves

RD	Status	Evidence (checked by the script)	Still open
RD-01	closed	F-03 in the rev 3 piece list: False	nothing (v3.1 set stays superseded with its STOP notice)
RD-02	restated	B-24 x2 + 2 carabiners in the meat-trip kit BOM: True; PT-6 in the release gates	user guide (RD-22) with the 53 kg limit and belly rule; PT-6
RD-03	closed	IM1 cut 990 mm in the rev 3 cut list	MT-1 / MT-2 on the sewn straps
RD-12	restated	B-17, B-18, B-22, HE, T3, LLT, TG drafted and sequenced: True; B-19 / B-20 / B-23 / T2 removed (BL-004 REC); IM3 hooks are wiregate carabiners (BL-010)	PT-1, PT-2, PT-3, PT-5, PT-6, LL-H before any load point is sewn on a bag that will carry
RD-15	closed (numbers)	BOM, yardage and mass use the current records (robic 125 gsm, ultragrid 132 gsm 1500 mm roll, ULTRA 133 gsm)	physical: swatch weights (QC-02), FS-O5 / PT-3 on 200X (alternate gate)
RD-16	closed (release)	CORRIE F rev 3 patterns (part A) + this op sequence replace SPINE step 9	first article build; CORRIE gates; BL-033 items listed as volatile ops
RD-17	restated	IMP-2 / IMP-4 / IMP-8 in the bag: True; SH-03 absent: True; body seams un-bound (SSa2, ASSUMPTION NO-BIND)	FRAY-1 wash-swatch test; IMP-1/3/5/6 belong to the harness / belt packages
RD-18	closed	B-11 x8 + BRL x2 + BRS + G-hook in the BOM: True	B-11 placements are not drawn on any rev 3 piece (pattern-maker)
RD-21	restated	this op sequence supersedes BUILD_PLAN.md step 9 for the bag (bag ops only)	BUILD_PLAN.md still describes stubs, bay mode, the PA66 plate: needs a pointer to this package (not edited here)
RD-22	open	the packaging card (README Packaging) carries the load limits	the user guide itself
RD-23	restated	B-25, B-27f, B-22 inside, TG, LLT drafted and sequenced (ops 3.04-3.05, 4.03): True	T9 rev 3 at the S kit; LL-H; PORT-1; attorney review; BL-033 may move B-22 (x +/-130.3 vs 149)
RD-24	restated	B-29 (7 piece types) + B-28 per kit + keepers sequenced; lid 9.53 h on the drafted pattern vs 7.76 h booked	T9 lid weigh / photo, cinch, free lengths; build_metrics still books the old lid hours (ledger / corrie_f owner)

12. Assumptions that govern

ID	Assumption	Settled by
NO-BIND	Body seams are SSa2 (second row over the turned allowances), not bound: the ledger books no binding (RD-17 B-1).	FRAY-1 wash-swatch test (3 cycles); if it fails, bind with 20 mm grosgrain via a binder
NO-TAPE	No taped seams: the bag is water resistant, contents in the user's dry bag.	owner decision if a waterproof bag is wanted (seam-tape machine + taped ULTRA seams)
SAM-RATIO	Factory SAM = 0.12 x beginner hours (bag_concepts.HOURS basis).	time study on the first 10-bag lot
LABOUR	Loaded CMT minute: US contract $0.85/min, Asia $0.12/min; overhead + margin 25%.	two contract quotes from this tech pack
THREAD-FACTOR	2.75 m of thread per m of 301 stitching (both threads, ends).	weigh the cones before / after the first bag
SPI-LIGHT	10 SPI on 40D-100D and mesh.	QC-05/06 swatch: no puckering, no perforation
THREAD-NEEDLE	Tex 70 / Nm 90, tex 90 / Nm 110, tex 135 / Nm 140, round point (attach_system.NEEDLE for 90 / 135).	swatches in op 0.02: no skipped stitches, no shredding
UTIL-SMALL	Small and sheet pieces nest at 70% across bags in production.	first production markers
UTIL-ALT	Alternate-set body fabrics nest at 85% in production (no 10-bag alternate marker).	alternate 10-bag markers if the alternate set passes its gates
B28-FACE	B-28 front tabs go on the OUTSIDE of the hinge band (the strap runs on the aft face); part A README says 'inside'.	pattern-maker / pack-designer confirm; T9 strap path
B25-BOX	Each B-25 toggle tab and second fold is closed with a 21 x 20 box-X (pattern shows folds only).	LL-H toggle-loop pull
IM-BOX / US-BOX / W3-BOX	IM1-IM3 end loops, US eyes and W3's hook end get a 21 x 25 box-X (pattern draws folds only).	MT-1 specimens are sewn exactly this way
WX-LOAD	Haul extenders carry the S9 float-clamp load (225 N).	pack-designer load case; LL-H
KW-PLACE	K-W keepers at W1-W3 heights z 80 / 250 / 420 in the aft corner seams (not drawn).	pattern-maker notches
F02-PLACE	F-02 centred on the outside of B-05 (not drawn).	pattern-maker placement; fly fit (BL-033)
BT-STD	Slit-end bartack 10 x 2 mm, 28 stitches; standard tack 16 x 2 mm, 42 stitches (LK-1900 patterns).	QC-07
PRICES	Items with no record cost use PRICE_ASSUMED (sliders, cams, tri-glides, toggles, elastic, labels).	supplier quotes; materials-engineer records
ELASTIC	25 mm elastic keepers: no material record (pattern module ELASTIC).	materials-engineer record